Geochemistry of sphalerite from Pine Point, NWT, Canada

by

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Abstract

Critical metals have become indispensable in the transition towards clean and sustainable energy. The Pine Point Pb-Zn district, located in the North-west territories of Canada has been described as a low temperature MVT deposit. These are known to host significant concentrations of critical metals such as gallium (Ga), germanium (Ge), and indium (In). Prior research has identified that Ga and Ge tend to be concentrated in low-temperature deposits, whereas In is predominantly associated with high-temperature environments. A study on the concentrations of these elements in various deposits indicated enrichment of In at Pine Point; however, the factors influencing its enrichment and distribution remain unclear.

In recent years, the laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) has become an indispensable tool for the investigation of trace element concentrations. In this study, we employ the use of the electron microprobe and the LA-ICP-MS to quantify the concentration of critical metals at Pine Point and to identify the potential geological controls influencing their distribution. Sphalerite at Pine Point is enriched in Fe, Cd, Mn, Pb, Ga, and Ge, most of which show smooth ICP-MS signals, indicating their presence as solid solutions in the mineral. Elements such as Cu, Co, Ag, and Tl are most likely present as mineral inclusions, and little to no traces of In was found at Pine Point. Early, colloform sphalerite is generally enriched in gallium, while germanium shows a preference for both colloform and later formed, iron-rich, coarse sphalerite. We posit that Ga prefers early, colloform sphalerite, possibly due to rapid sulphide deposition associated with this texture. Correlation plots show a slight positive correlation between Ge and Fe (0.26), suggesting a coupled substitution of $Ge^{4+} + 2Fe^{2+} + \text{vacancy} \leftrightarrow 4Zn^{2+}$.

Preface

This thesis is an original work by Michelle Solomon Woje. No part of this thesis has been previously published.

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1. Introduction

The growing demand for critical metals has sparked an interest in Mississippi Valley-type (MVT) deposits, which host significant quantities of these metals. Metals such as indium (In), gallium (Ga), germanium (Ge) are considered critical by many countries including the government of Canada (The Canadian Critical Minerals Strategy). They are fundamental for the transition to green energy as they play a vital role in the manufacture of modern technology and are essential to many industries (Jowitt et al., 2018). Gallium in the form of gallium arsenide (GaAs) and gallium nitride (GaN) is used to manufacture light emitting diodes (LEDs), integrated circuits (ICs), which have applications in the aerospace industry, and the production of medical equipment. (Jaskula, 2024). Germanium is employed in the production of fibre optic cables and solar cells. (Tolcin, 2024). Indium has applications as a thin-film coating for electronic devices, such as alloys and electrical components. (Stewart, 2024). As the world shifts towards more green technologies to minimize their reliance on fossil fuels, the demand for these metals is expected to rise. (Paradis, 2015).

These critical metals are largely hosted in sphalerite ore (ZnS), a common mineral found in MVT deposits. The trace elements found in sphalerite have been used by previous authors to distinguish different deposit types, as ore deposits have geochemical signatures that differ resulting from differences in enrichment values of these elements. This is mostly owing to sphalerite's propensity to integrate trace elements, which is influenced by variables such as ore formation temperatures, formation mechanisms, fluid and metal sources, and fluid evolution. Frenzel et al., (2016) proposed the GGIMF (Ga, Ge, In, Mn, Fe) geothermometer using the trace element signatures and homogenization temperatures of sphalerite in different deposits as a means of studying the formation conditions of sphalerite in different Pb-Zn deposits. Some authors have linked high indium concentrations with high temperature, magmatic-hydrothermal deposits; and high gallium, germanium concentrations with low temperature deposits. (Cook et al., 2009; Belissont et al., 2014; Frenzel et al., 2016; Li et al., 2023).

Paradis (2015) noted elevated levels of indium at Pine Point, Northwest Territories, Canada — a low temperature MVT deposit. As previous studies show a general trend of higher indium concentrations associated with high temperature deposits, this anomaly raises important questions about the geochemical processes involved in the enrichment of critical metals at Pine Point, as well as the concentration and distribution of these metals. Previous studies have been carried out on Pine Point focusing on the geology, paragenetic sequence, its potential to produce lead and zinc, and the mechanism of precipitation (Krebs et al., 1984; Adams et al., 2000; Paradis et al., 2005; Hannigan, 2007; Szmihelsky et al., 2020). Nevertheless, limited research has employed laser-ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) to investigate its potential as a host for critical metals and to examine the factors influencing enrichment. By examining the mineralogical and geochemical characteristics of the deposit, we aim to explore the concentration of critical metals at Pine Point, analyse the patterns, understand the controls on distribution and propose explanations for this anomalous concentration of indium. This study utilized electron microprobe analysis (EPMA) and LA-ICP-MS analysis to determine the major and trace element compositions to provide insights into the conditions that favour the concentration and enrichment of these metals.

2. Geology of the Pine Point deposit

The Pine Point mining district is Canada's most significant MVT deposits. Located on the south side of Great Slave Lake in the Northwest Territories, the district is made up of about 100 known Pb-Zn orebodies (Hannigan, 2007). These are hosted in Devonian platform, carbonate rocks of the Presqu'ile Barrier Reef, which overlay the Archean basement complex (Rhodes et al., 1984) and is comprised of the Watt Mountain, Sulphur Point, and Pine Point Formations (Figure 1). The Presqu'ile Barrier Reef has a thickness of about 200 m around the Pine Point area and separates the open sea, MacKenzie shale basin to the north from the Elk point evaporite basin to the south. Pine Point is in the eastern part of the barrier (Jackson et al., 1969, Qing, 1998). The basal part of the Pine Point stratigraphy is comprised of early, Givetian, Keg River Formation made up of dolostones, limestones with brachiopod fragments (Figure 2). The barrier reef is made up of the dolomitized limestones of the Pine Point Formation as well as the dolomitic limestone, and sparry calcite of the overlying Sulphur Point Formation. These formations grade towards the crystalline dolomite and anhydrite of the Muskeg Formation, deposited in the Elk Point Basin (Qing et al., 1994). The micritic limestones and dolostones of the Watt Mountain Formation overlie the Sulphur Point Formation unconformably to the south of the barrier complex and the Windy Point Formation to the north (Skall, 1975; Rhodes et al., 1984). Mineralization at Pine Point resulted from basinal brines with temperatures ranging from 50-100ºC (Kyle,1981; Roedder, 1968). Some authors posit that the McDonald fault (Figure 1), part of the Great Slave Lake shear zone may have assisted in transmitting the fluids responsible for mineralization at Pine Point (Skall, 1975; Krebs and Macqueen, 1984). The source of base metals is not certain, but sulphur was most likely sourced from brines of the Elk point basin (Szmihelsky et al., 2020). Mineralization at Pine Point is epigenetic (Kyle,1981) and occurred as a result of the mixing of brine with oil which triggered the deposition of sulphides (Szmihelsky et al., 2020). Ore composition at Pine Point includes sphalerite, galena, marcasite, with some pyrite. Gangue minerals include dolomite, calcite, celestite and native sulphur (Kyle,1981).

2.1 Textures and orebody morphology

Orebodies at Pine Point vary greatly in geometry, texture, and size. Previous authors have classified these orebodies as tabular, prismatic or abnormal based on their vertical and horizontal dimensions (Kyle 1977, 1981; Skall, 1975; Cumming et., al 1990). Rhodes et al. (1984) cite karstification as the primary control on ore deposition at Pine Point. They show a substantial correlation between the geometry of the karst structure and that of the matching orebodies they host. They describe four orebody types: the widespread, elongate, flat-lying orebodies of the tabular karst; the vertically elongate orebodies of the prismatic karst that resulted from extreme karstification in the tabular karst ore horizon; the abnormal prismatic orebodies that resulted from the mineralization of unconsolidated collapse structures; and the mineralized, subhedral crystals with fossil vuggy porosity of the B spongy type. In this paper, we describe tabular orebodies as having greater horizontal than vertical dimensions, prismatic orebodies as having greater vertical than horizontal dimensions, and abnormal orebodies as those that do not fall into any of the two aforementioned categories (Figure 3).

3. Sampling and analytical methods

Fourteen representative samples were selected from ten orebodies that encapsulate the different sphalerite textures and the different orebody geometries recognized at Pine Point. Selected samples represent tabular orebodies with coarse texture and some with colloform texture; prismatic orebodies with coarse texture and some with colloform texture; and some abnormal orebodies. Samples were selected from the W17, M40, N32, O42, K62, K57, O28, N38, N204, X15 orebodies (Table 1).

3.1 Petrography

Samples were studied under transmitted and reflected light microscopy using an Olympus BX53M optical microscope at the University of Alberta. Textural and mineralogical characteristics of minerals were examined to establish a paragenesis. This was used to select representative samples from a greater sample size.

3.2 SEM imaging

Imaging was carried out at the University of Alberta's Scanning Electron Microscope lab using a Zeiss Sigma Field Emission SEM equipped with secondary and backscattered electron detectors, and a Bruker energy dispersive X-ray spectroscopy (EDS) system. SEM-BSE imaging was used to investigate textural and paragenetic relationships at 15kv accelerating voltage.

3.3 EPMA

Major elements were analysed at the Electron Microprobe Analysis Laboratory (Department of Earth Science, University of Alberta) using a CAMECA SX100 equipped with five wavelength dispersive spectrometers (WDS), and an energy dispersive spectrometer (EDS). Samples were carbon coated (25 nm thick) prior to analysis. Major, and trace elements were measured with a beam current of 30 nA, an accelerating voltage of 20 kv, and a beam diameter of 2 μm. Measurements were performed using wavelength dispersive spectrometry (WDS) and Probe-for-EPMA software (Donovan et al. 2015), and corrections applied to account for dead-time effects (Donovan et al. 2023). WDS elemental distribution maps were collected on the same instrument at 20 kv accelerating voltage, 100 nA beam current, and 5 μm beam-size. The reported elements of sphalerite include Zn, Cd, Pb, Fe, Cu, S, and As, and the data was corrected with the ZAF correction program. Zinc concentrations of sphalerite were used as an internal standard for LA-ICP-MS analyses.

3.4 LA-ICP-MS analyses

The trace element content of sphalerite was analysed by an NWR 193 UC laser-ablation system coupled to an Agilent 7900 quadrupole mass spectrometer at the University of Toronto. Test runs on samples were carried out on the first day with 30 seconds background measurement, 30 seconds analysis time, and a 30 second wash-out delay. Laser ablation was carried out at 5 Hz pulse rate and a laser fluence of 4.5 J/cm^2 was used. The laser beam size was set to 30 µm, scan speed at 10 µm/s. Subsequent analysis was conducted with 30 seconds background measurement, 45 seconds analysis time, and a 30 second wash-out delay after analysis for each spot analysed. To monitor and correct for instrumental drift, reference materials were analysed initially and re-analysed after every 20 spots of the unknown samples. Nineteen elements were measured in total namely: ^{25}Mg, ^{34}S, ^{55}Mn, ^{57}Fe, ^{59}Co, ^{60}Ni, ^{63}Cu, ^{66}Zn, ^{71}Ga, ^{74}Ge, ^{75}As, ^{107}Ag, ^{111}Cd, ^{115}In, ^{118}Sn, ^{121}Sb, ^{202}Hg, ^{205}Tl, and ^{208}Pb. GSD-1, STDGL3, and Sph were used as Reference Materials to correct for instrument drift, and to provide quantitative trace element values; and the MASS-1 standard developed by Wilson et al. (2002) was used as an external standard. The data collected was processed using the Iolite version 4 software (Paton et al., 2011). Detection limits were calculated using Longerich et al. (1996) and were generally below ppm concentration for the trace elements analysed.

4. Results

4.1 Paragenesis

Optical microscopy, and SEM techniques were employed to outline a paragenetic sequence at Pine Point. Figure 4 provides an overview of the paragenesis, which is further detailed below. The paragenesis at Pine Point varies in detail from one orebody to the next, but some general trends are observed. Mineralogy at Pine Point is consistent with that of other MVT deposits and includes dolomite, calcite, sphalerite, galena, pyrite, and marcasite. The mineral assemblage at Pine Point is simple and follows the sequence outlined in Kyle, 1981 shown in Figure 4. It is characterized by open space filling textures, and common directional growth indicators, which help to clarify the sequential growth of minerals. Dolomite is brown or grey (Figure 5a) and is earliest with occasional traces of pre-ore calcite. Early calcite appears white in hand samples and opaque in transmitted light (Figure 5g, h). Saddle dolomite often forms after euhedral crystals of brown-grey dolomite predating sulphide precipitation but is sometimes seen postdating ore mineralization. Marcasite is the earliest and is commonly intergrown with sphalerite or overgrown by blocks of pyrite (Figure 5c, e). Sphalerite is the most abundant sulphide. There are two principal textures of sphalerite common at Pine Point, which represent two generations of growth. Colloform sphalerite is early and characterized by micro-crystalline textures and rhythmic banding with alternating colours that range from clear, yellow, red, dark brown, to black (Figure 5a). Coarse sphalerite, forms after and has well- formed colour zones with darker, iron rich sphalerite typically earlier, followed by lighter, iron-poor sphalerite. However, this sequence is not continuous and forms alternating bands of light and dark sphalerite in some samples (Figure 5d). Sphalerite colours range from pale, yellow, red to dark brown; this change in colour is largely due to increasing iron concentrations. Galena is usually the last sulphide precipitated and takes a blocky, cubic or dendritic form Figure 5f. It is most times coeval with sphalerite or found overgrowing sphalerite. Dendritic forms of galena are commonly associated with early, colloform sphalerite as described in Fowler et al., 1996 (Figure 5i). Later formed cubic, blocky galena is intergrown with coarse sphalerite textures. Post-ore calcite is the latest mineral formed and is often seen as coarse grains filling up fractures and pores within sulphides or being coeval with later formed saddle dolomite (Figure 5j).

4.2 EPMA

The data collected from the microprobe is summarised in Appendix A, with means, maximum and minimum values, and standard deviations for each sample, WDS maps are shown in Figure 6. Zinc values range from 52 to 67 wt% and vary mostly with iron enrichment. This is represented in the colour changes observed. Iron values range from 0.04 to 14 wt%, with highest values typically seen in reddish-brown zones of sphalerite (Figure 6), and less iron in lighter, pale-yellow, to transparent sphalerite. However, darker bands in colloform sphalerite do not necessarily equate to high iron content as described by Roedder and Dwornik (1968). Sulphur values are relatively constant and range from 30.2 to 33.5 wt%. Lead is the second most predominant minor element, after iron with values ranging from 0 to 1.3 wt%. Notably, Pb content exhibits no discernible correlation with sphalerite colour.. Cadmium concentrations range from 0 to 0.67 wt%, while copper and arsenic had negligible values.

4.3 LA-ICP-MS trace and minor element concentrations

Trace element concentrations measured by LA-ICP-MS are given in Appendix B. Sphalerite at Pine Point hosts a variety of trace elements; Fe, Cd, Mn, Cu, Ga, Ge, Pb, Co, Hg, Tl, Ag, Ni, Sb were present in significant quantities above detection limits for most samples and vary considerably. As, In, Sn had mostly negligible concentrations and were mostly below the minimum detection limit. *Manganese* values range from 0.3 to 1065 ppm with a mean value of 89 ppm. The concentration of Mn is consistent with values reported for MVT deposits (Ye et al., 2011) and aligns with the observed positive correlation between Mn and Fe. *Iron* is the most abundant minor element in sphalerite with concentrations ranging from 38 ppm to 34800 ppm, and an average of 2460 ppm. Dark, red zones of colloform sphalerite show the highest Fe concentrations. *Cobalt* values are generally unevenly distributed. Values range from 0.01 ppm to 45 ppm, with an average of 1 ppm. These elevated values are typically seen in samples with very high Fe, suggesting a positive correlation between the two elements. The N204 sample shows the highest consistent values for Co. *Nickel* follows the trend of Co concentration, with the highest concentrations in very Fe-rich sphalerite that contains high values for Co. This suggests a positive correlation between Ni, Co, and Fe. Ni values range from detection limits to 53 ppm, with an average of 0.7 ppm. *Copper* concentrations are highly variable, with most values less than 1 ppm. Values range from 0.01 ppm to 75 ppm, with an average value of 1 ppm. These values are low compared to Cu values described for MVT deposits by (Frenzel et al., 2016, Cook et al. 2009; Yuan et al. 2018) and suggest the presence of micro-inclusions. Cu follows a general trend of being enriched in very Fe-poor, light coloured, coarse sphalerite. *Gallium* concentrations are consistent with values described for MVT deposits by (Cook et al. 2009; Yuan et al. 2018). They are generally low with above average values concentrated in dark brown, colloform, banded sphalerite and the lowest values in coarse sphalerite. Values range from detection limits to 89 ppm, with an average value of 3 ppm. *Germanium* has higher concentrations when compared to gallium, consistent with values reported in MVT deposits (Cook et al. 2009; Ye et al. 2011, 2016; Yuan et al. 2018). It shows a more even distribution across samples and signals are generally smooth (Figure 7). Germanium concentrations vary from 0.07 ppm to 614 ppm, with an average value of 44 ppm. Significant Ge enrichment is typically accompanied by high concentrations of Fe. Highest Ge concentrations reported are in coarse sphalerite. *Silver* values are usually below 1 ppm but evenly distributed across all samples in measurable quantities. Time resolved signals however show an irregular profile (Figure 7) with values ranging from 0.4 ppm to 2 ppm, and a mean of 0.7 ppm. *Cadmium* distribution across sphalerite shows a negative correlation with Fe as has been noted by previous authors (Li et al., 2020; Liu et al., 2022). Cd values are lower than expected for MVT deposits described by (Cook et al. 2009; Frenzel et al., 2016; Yuan et al. 2018). Concentration ranges from detection limits to 20 ppm, with an average value of 1.7 ppm. Despite the low values, Cd is widespread across all samples and shows smooth signals for most samples (Figure 7). *Indium* values were generally below 1 ppm, with the highest In value coming up to 0.004 ppm. *Tin* concentration is negligible at Pine Point, like other MVT deposits. Values range from detection limits to 5 ppm, with an average of 0.1 ppm. This partial enrichment is seen in Sample N204, an abnormal orebody that hosts both micro-crystalline and coarse sphalerite textures. This anomalous concentration is not specific to Sn alone as Co, Ga, Fe, Ni, Mn, Cu, Cd, Sb have very high values present in this sample. *Mercury* values are generally less than 1 ppm and vary from detection limits to 15 ppm, with an average of 0.3 ppm. Time resolved signals for Hg show a smooth and consistent signal with sample K62 showing consistent values of Hg above 1ppm. High values of *Thallium* are sometimes related to Pb enrichment, as both elements are most times concentrated in the same zones, with a value range from detection limits to 4 ppm, and a mean concentration of 0.4 ppm. *Lead* is the second most abundant minor element at Pine Point with concentrations within the range of 0.2 ppm to 5360 ppm and an average value of 282 ppm. High values of Pb are generally noted in dark cores of colloform sphalerite, with the highest values recorded in the M40-176 sample. Pb signals have irregular profiles for some samples, indicating the presence of mineral inclusions, probably galena.

5. Discussion

5.1 Distribution of trace elements and genetic type

The colour of sphalerite has been linked to its iron content, primarily due to the substitution of Fe^{2+} for Zn^{2+} within the crystal lattice, as documented by previous studies. (Wei et al., 2021; Wang et al., 2023). Additionally, some have proposed that trace elements affect the colour of sphalerite. Pfaff et al., 2011; Gagnevin et al., 2012 correlated the colour changes in colloform sphalerite with As content. Arsenic values at Pine Point were generally low, and sphalerite colour does not appear to be influenced by As concentrations. Figures 8, 9, and 10 show a marked correlation between Fe content and sphalerite colour, with the highest Fe concentrations in early, dark brown sphalerite and significant depletions in light to transparent sphalerite. This correlation does not extend to black variations of sphalerite as Liu et al., 2023 noted. Black colloform sphalerite in the O42 sample shows lower values of Fe and other trace elements, suggesting a different control on the very dark colour of colloform sphalerite (Figure 8). Pb would be a plausible control, however, Pb values here are not significant and the ICP-MS signals suggest the presence of mineral inclusions to account for Pb values (Figure 7c). The general trend of Fe depletion at Pine Point progresses from early to late (Figures 8, 9, and 10) — nevertheless, the presence of alternate bands of light and brown coarse sphalerite suggests many generations or pulses of Fe-rich fluids, rather than a singular mineralizing event (Figure 5d). No correlations were observed between high values of Fe and the presence of iron sulphides.

Previous studies have described the trace element signature of sphalerite as a function of deposit type. Elevated levels of indium are commonly associated with Fe, Cu, Ag, Sn, and have been linked to high temperature deposits, such as SEDEX, skarns, and VMS. These deposits are notably depleted in Ge, Ga. Low temperature ores, such as MVT deposits are characterized by Cd, Ga and Ge enrichment. (Cook et al., 2009; Ye et al., 2011, Shimizu et al., 2012; Cook et al., 2015; and Frenzel et al., 2016). Sphalerite at Pine Point is characterized by measurable concentrations of Ga, significant concentrations of Ge, and little to no concentrations of In, with most In values being below detection limit. Notably, some of the other elements commonly associated with In enrichment are very low in concentration at Pine Point. Consistent concentrations of Ga above the average value of 3 ppm were reported in early, colloform sphalerite (Figures 8, 9, and 10). This suggests that the early fluid was enriched in Ga. Another explanation for these concentrations can be attributed to the growth rate of colloform sphalerite, as colloform textures are a general indication of supersaturation and rapid deposition (Kyle,1984; Shimizu and Morishita, 2012). At such conditions, Ga would be precipitated and enriched in colloform sphalerite. Although the highest Ga values were reported in the N204 abnormal orebody, there is no relationship between Ga content and orebody type. The N204 sample shows inclusions present in the sample (Figure 7f), that could explain the anomalous concentrations. Ge values at Pine Point are distributed across the different textures with the highest consistent values present in coarse sphalerites of the prismatic K57 and O28 orebodies (Appendix B), the W17 orebodies are however, very low in Ge. This suggests a preference for coarse sphalerites of the prismatic orebodies. The colloform textures of the X15 orebodies are also very Ge-rich. Although MVT deposits are characterized by elevated Ga, Ge concentrations, no correlation was found between the two elements at Pine Point (Figure 11e). Cadmium values at Pine Point are lower than expected — nevertheless, the relatively low temperatures, high Ge, and low In are consistent with typical MVT deposits.

5.2 Trace element substitution mechanisms

The presence of trace elements in sphalerite is also controlled by substitution mechanisms into the sphalerite lattice. Divalent cations such as Cd^{2+}, Cu^{2+}, Fe^{2+}, and Mn^{2+} can directly substitute for Zn^{2+} (e.g. $Zn^{2+} \leftrightarrow Fe^{2+}$, $Zn^{2+} \leftrightarrow Mn^{2+}$) due to their similar ionic radii and charge. (Cook et al., 2009; Ye et al., 2011; George et al., 2016). For trivalent and tetravalent cations (Ga^{3+}, In^{3+}, Sn^{3+}, Ge^{4+}, Sn^{4+}), a coupled substitution involving monovalent ions (Ag^{+}, Cu^{+}) is proposed (Cook et al., 2012; Belissont et al., 2014, Wei et al. 2019). Cd, Fe, Mn, Ge, and Pb show smooth ICP-MS signals suggesting their presence as solid solutions in sphalerite and possible substitutions for Zn. Fe is the most abundant minor element at Pine Point; therefore, correlation plots can be used to determine whether the incorporation of other elements in the sphalerite lattice is facilitated by the presence of Fe. Fe shows a strong positive correlation with Mn, a weak correlation with Pb, and a negative correlation with Cd with correlation coefficients (R^2) of 0.73, 0.21, and 0.16 respectively (Figures 11b, c, and d). This suggests that these elements substitute directly for Zn in sphalerite.

Previous studies have highlighted several coupled substitution mechanisms for Ge into sphalerite. Coupled substitutions for Ge involve Cu or Ag, which are often enriched alongside Ge and plotted with to show correlations. Yuan et al. (2018) suggested a coupled substitution for Ge and Fe ($Ge^{4+} + 2Fe^{2+} + \text{vacancy} \leftrightarrow 4Zn^{2+}$) due to the correlation observed between the two elements at the Daliangzi Pb–Zn deposit in China. Ge can be substituted directly into the sphalerite lattice as $Ge^{2+} \leftrightarrow Zn^{2+}$ or as $Ge^{4+} + \square \leftrightarrow 2(Zn^{2+}, Fe^{2+})$ ($\square$ = vacancy). It can also be incorporated into sphalerite as $Ge^{4+} + 2Cu^{+} \leftrightarrow 3Zn^{2+}$, $Ge^{4+} + 2Ag^{+} \leftrightarrow 3Zn^{2+}$ (Belissont et al., 2016; Cook et al., 2009, 2015; Belissont et al., 2014; Cugerone et al., 2021; Wei et al., 2021). Correlation plot of Fe vs Ge shows a weak, positive correlation (0.26) between the two elements (Figure 11a). Other substitution mechanisms were explored, however, Ge at Pine Point showed a negative correlation with Cu and Ag (0.10, 0.01 respectively) as these elements are poorly concentrated with ragged ICP-MS signals (Figure 7). Luo et al., 2022 and Sun et al., 2023 proposed more complex substitutions involving Mn and Pb. Our data shows smooth signals for these elements, however — correlation plots of Ge against these divalent cations showed very little correlation (0.18, 0.11 respectively) (Figure 11 f, g), eliminating $Ge^{4+} + M^{2+}$ + vacancy $\leftrightarrow$ $3(Zn,Cd)^{2+}$, where (M = Pb, Mn) as a possible coupled substitution mechanism. The Pine Point sphalerite does not seem to follow any substitution mechanism that has been proposed so far. Although, a reduced form of Ge (Ge^{2+}) in sphalerite has been proposed by Bonnet et al. (2017), this form of Ge lacks significant empirical evidence and is poorly documented in the literature. Other authors (Frenzel et al., 2016; Bauer et al. 2019) have proposed Ge enrichment as a factor of low temperature. Thus, based on the observed correlations between Ge and other elements, we propose: (1) a coupled substitution as ($Ge^{4+} + 2Fe^{2+}$ + vacancy $\leftrightarrow 4Zn^{2+}$); (2) a direct substitution as ($Ge^{4+} + \square \leftrightarrow 2(Zn^{2+}, Fe^{2+})$) ($\square$ = vacancy) yielding a site vacancy; (3) the influence of temperature and/or fast mineral growth rate to explain Ge incorporation into the sphalerite lattice.

The ubiquitous form of Ga in sphalerite occurs as Ga^{3+}, which is incorporated into the sphalerite lattice as coupled substitutions in the form of $(Cu, Ag)^{+} + Ga^{3+} \leftrightarrow 2Zn^{2+}$. (Bonnet et al., 2016, Wu et al., 2024). Ga shows smooth profiles for some spot analyses and ragged profiles for others, indicating its presence as a solid solution and as possible mineral inclusions. Correlation plots of Ga show weak positive relationships with Ag and Cu (0.10 and 0.14 respectively). The coupled substitution involving $(Cu + Ag)^+$ shows a weak positive correlation to Ga as well (0.16) (Figure 11h) — nevertheless, it remains the most plausible substitution mechanism for Ga in sphalerite. However, we propose that rapid mineral growth from Ga-bearing fluids presents an even more likely explanation for its enrichment in sphalerite at Pine Point.

5.3 Ore-forming temperature, physicochemical conditions of the ore forming fluid

The changes in sphalerite texture from colloform to coarse, as well as variations in concentrations of trace elements are indicative of some form of transition in the mineralisation process and the conditions present at mineralisation. Previous authors have reported correlations between Mn enrichment in sphalerite and reducing environmental conditions at the time of formation. (Wei et al., 2018, 2021; Yu et al., 2019; Zhuang et al., 2019). Colloform sphalerite textures have been associated with strongly reducing conditions, typically resulting from the mixing of brines with H_2S-rich fluids (Beales, 1975; Fowler et al., 1996; Niu et al., 2024), including at Pine Point itself (Adams et al., 2000). In contrast, coarse sphalerite at Pine Point has been linked to sulfate reduction processes, potentially driven by the mixing of brine with hydrocarbons (Szmihelsky et al., 2020), which may reflect slightly less sulfidic but still reducing fluid conditions. Early, dark brown colloform sphalerite is enriched in Mn (up to 1065 ppm), consistent with strongly reducing conditions where Mn^{2+} is stable and mobile. However, Mn is significantly depleted in some zones of early, coarse, white sphalerite before concentrating again, suggesting a shift in fluid chemistry—possibly to conditions with less reducing potential or lower Mn availability.

These variations suggest multiple pulses of mineralization, where redox conditions fluctuated between strongly reducing (H_2S-rich) and moderately reducing (sulfate-reducing). The possibility of a reducing ore fluid being transported to the site of deposition periodically must be considered. Precipitation mechanisms for the different textures of sphalerite changed from early to late. Luo et al., 2022 and Liu et al., 2023 also noted these changes from colloform to coarse sphalerite, attributing them to changes in the growth rate, pH; and a decrease in oversaturation.

In addition, the composition of ore fluids influences the distribution of trace metals in sphalerite. Trace elements at Pine Point sphalerite contain variable concentrations of chalcophile (Cu, Pb, Zn, Ag, Hg, As, Cd) and siderophile (Fe, Ni, Co) elements. Generally, trace elements show a greater affinity for early, dark brown colloform sphalerite, compared to the later formed, coarse textures, indicating a heterogenous fluid composition at Pine Point. Fe, Mn, Pb, Ga, Ge show high concentrations in colloform sphalerite, while Cd generally tends to prefer Fe depleted areas. The negative correlation between Fe and Cd suggests that Cd is preferentially incorporated into the sphalerite lattice as the fluid becomes depleted in Fe, possibly through direct substitution. During the deposition of colloform sphalerite, ore fluids appear to have been enriched in Fe, Mn, Ga, Ge, Co, Ni and trace element composition decrease as deposition progresses. Sulfidation state is higher at this point compared to later formed textures, which are mostly rich in Ge, with occasional concentrations of Hg and Ag in the K62 sample (Appendix B). Pale to transparent sphalerite, which often forms last is mostly depleted in trace elements, with some level of Cd concentration, suggesting that the fluid responsible was highly depleted in Fe, and slightly concentrated in Cd. Even when coeval with galena, Pb concentrations are generally low here. Thus, the fluid composition demonstrates a major influence on the variability of trace element composition from early to late sphalerite.

Numerous studies have explored a sphalerite geothermometer that correlates trace element content with formation temperature, as trace elements are incorporated into the sphalerite lattice at varying temperatures (Keith et al., 2014; Frenzel et al., 2016; Frenzel et al., 2022; Zhao et al., 2024). The minimal concentrations of In at Pine Point preclude the application of the GGIMF calculator by (Frenzel et al., 2016). The SPRFT software created by (Zhao et al., 2024) was instead utilised in this study to estimate the temperature of sphalerite. This study utilised Mn, Fe, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, Sb, and Pb, as these elements have been commonly referenced in the literature due to their correlation with mineralisation temperature. Sphalerite enriched in Cu, Fe, Mn, In, and Sn suggests high temperature conditions of formation, whereas sphalerite formed at low temperatures is characterised by enrichment in Cd, Ga, and Ge (Cook et al., 2009; Ye et al., 2011; Bauer et al., 2019). Cao et al. (2014) delineated the characteristics of sphalerite establishing a correlation between colour to temperature. High temperature sphalerite was notably darker and enriched in Fe and Mn, while low temperature sphalerite was described as having a lighter colour and a relative depletion of these elements. Using these studies, we attempt to decipher the temperature of sphalerite using its trace element signature. At Pine Point, sphalerite is characterised by enrichment in both low and high-temperature elements such as Fe, Mn, and Ge; with colours ranging from black to transparent. This classifies Pine Point as a low-medium temperature deposit. Previous authors have used homogenisation temperatures to describe this deposit as low temperature, with temperatures ranging from 50–100 °C (Kyle,1981, Roedder, 1968). Ore forming temperatures calculated by the SPRFT software were in the range of 136-204 °C, with early colloform textures having higher temperatures (178 ± 26 °C) and later formed coarse textures with lower temperatures (162 ± 26 °C). This is consistent with the observed variations in sphalerite colour and trace element descriptions of (Cook et al., 2009; Ye et al., 2011; Cao et al., 2014) in relation to temperature. The calculated temperature range however exceeds the homogenisation temperatures previously documented for Pine Point. It can therefore be suggested that Pine Point falls within the range of low-medium formation temperatures.

5.4 Implications for exploration

Boxplots showing the distribution of minor and trace elements across the three distinct identified orebodies are presented in (Figure 12), with median values employed for comparative analysis. Overall, abnormal orebodies tend to have higher median values for minor and trace elements. Cd concentrations are similarly distributed across the different orebodies, with the highest values in Fe-depleted prismatic orebodies. Co and Cu have a wider spread in abnormal orebodies and better distribution in tabular orebodies. Fe, Ga, Ge and Pb have their highest median values in abnormal orebodies, while Mn shows high median values in tabular orebodies. Ge values show little variability across all orebody types. The correlation between Ge and Fe at Pine Point suggests that exploration for this critical metal should focus on orebodies where Fe concentrations are highest. All the data suggests that Pine Point contains substantial minor elements, with Ge being the most prominent critical element enriched, particularly in abnormal orebodies.

6. Conclusion

1. Sphalerite mineralisation at Pine Point is characterised by colloform and coarse textures found within various orebodies, including prismatic, tabular, and abnormal forms. Colloform textures tend to be paragenetically earlier than coarse textures. LA-ICP-MS analysis indicates that sphalerite at Pine Point contains a range of trace elements including

Fe, Cd, Mn, Cu, Ga, Ge, Pb, Co, Hg, Tl, Sb, Ag, and Ni, with variations observed across different texture types. Among them, Ge stands out as the most abundant of the critical metals of interest, while concentrations of In were mostly below detection limits. Colloform sphalerite that is rich in Fe seems to contain the greatest quantities of trace elements, such as Ge.

2. These minor elements often substitute for Zn in the sphalerite lattice. The positive correlation between Ge and Fe suggests that Ge is easily substituted into sphalerite through a coupled substitution with Fe ($Ge^{4+} + 2Fe^{2+} + \text{vacancy} \leftrightarrow 4Zn^{2+}$); while concentration of Ga is attributed to the mechanism of precipitation of the colloform textures it is enriched in.
3. We suggest that colloform sphalerite formed as a result of oversaturation, fluid mixing, and fast growth under reducing conditions. This process facilitated the concentration of the critical metals of interest, along with various trace elements within this sphalerite texture. Coarse textures indicate slower growth rates and tend to concentrate fewer trace elements, with the exception of Cd and Ge in certain cases. Trace element compositions of sphalerite place Pine Point as a low-medium temperature MVT deposit.
4. We posit that the abnormal and prismatic orebodies at Pine Point show the best prospect in terms of Ge exploration.

Tables

Tabular	Prismatic	Abnormal
M40.18. PP.135HT12	W17.44.6HT12	N204.ISA.77HT12
N32A-2F	O42-1	X15.28.02
M40.176.9/12HT12	W17.44.6HT11	X15.393-5HT11
	K62C	
	K57BFIT12	
	W17.46.203	
	O28-2	
	N38A-4	

Table 1: Classification of samples according to orebody types.

Figures

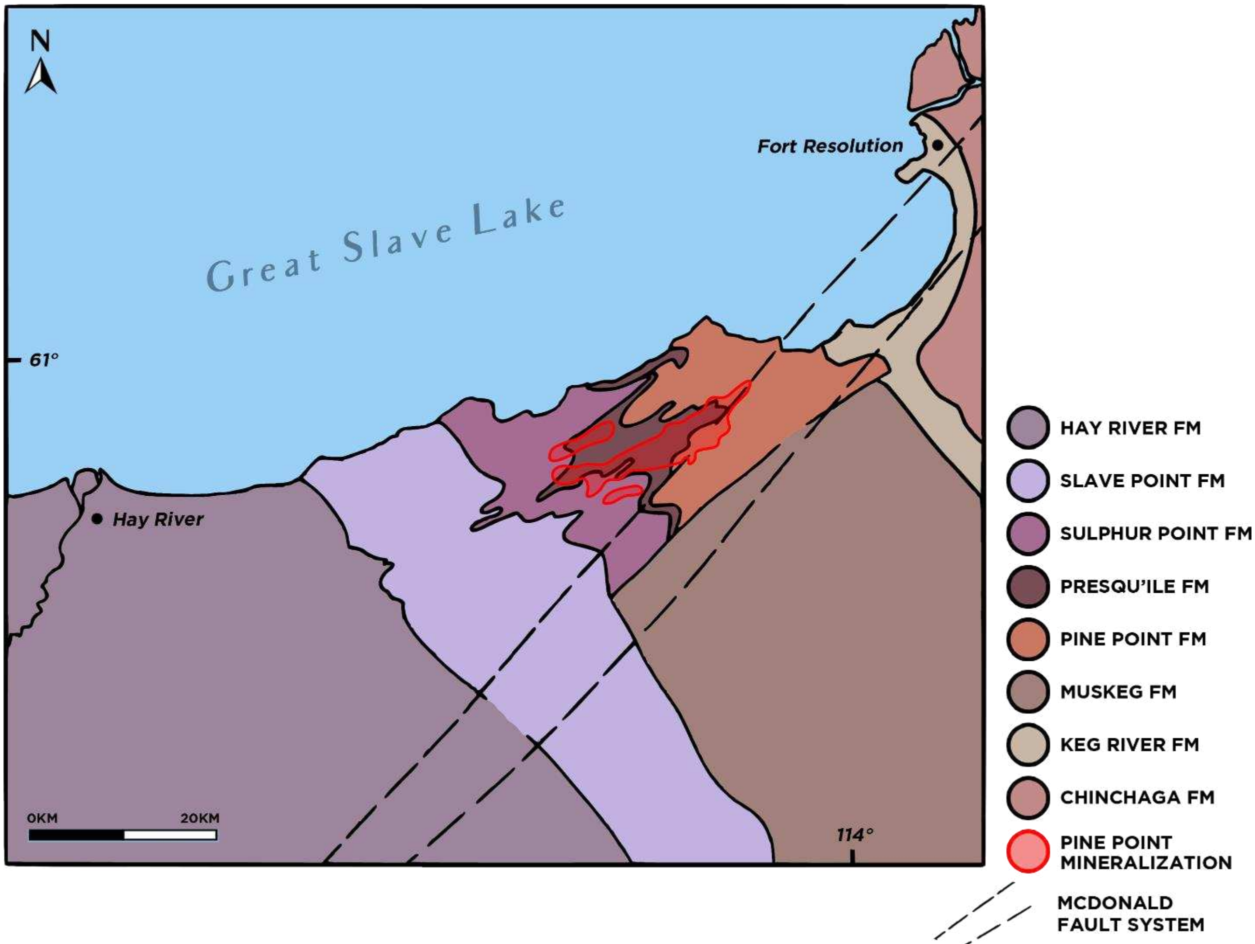


Figure 1: Geologic map of the Pine Point mining district (modified from Krebs and Macqueen, 1984 and Szmihelsky et al., 2021)

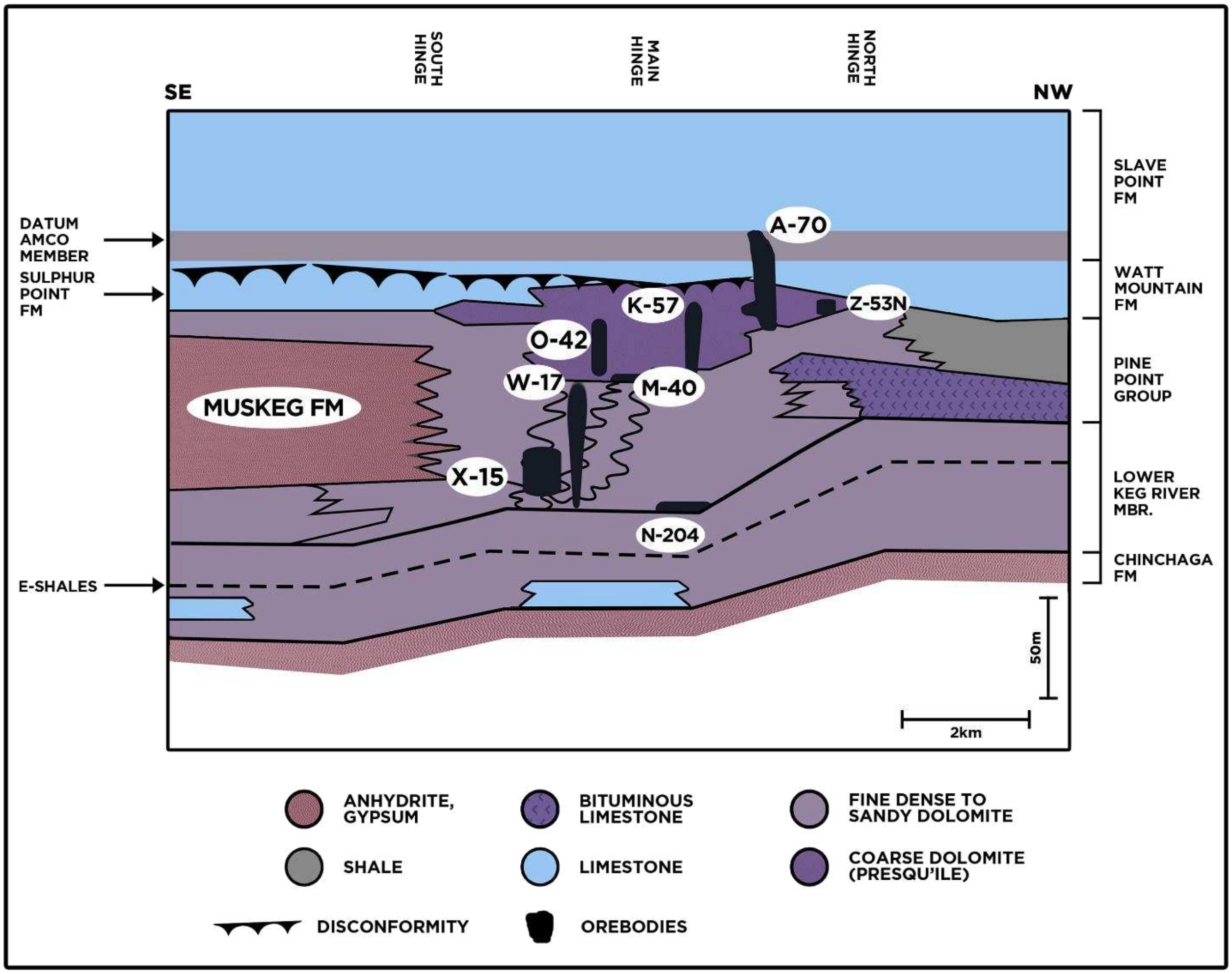


Figure 2: Cross section of the Pine Point mining district showing selected orebodies and their relationship to the stratigraphy (modified after Skall, 1975).

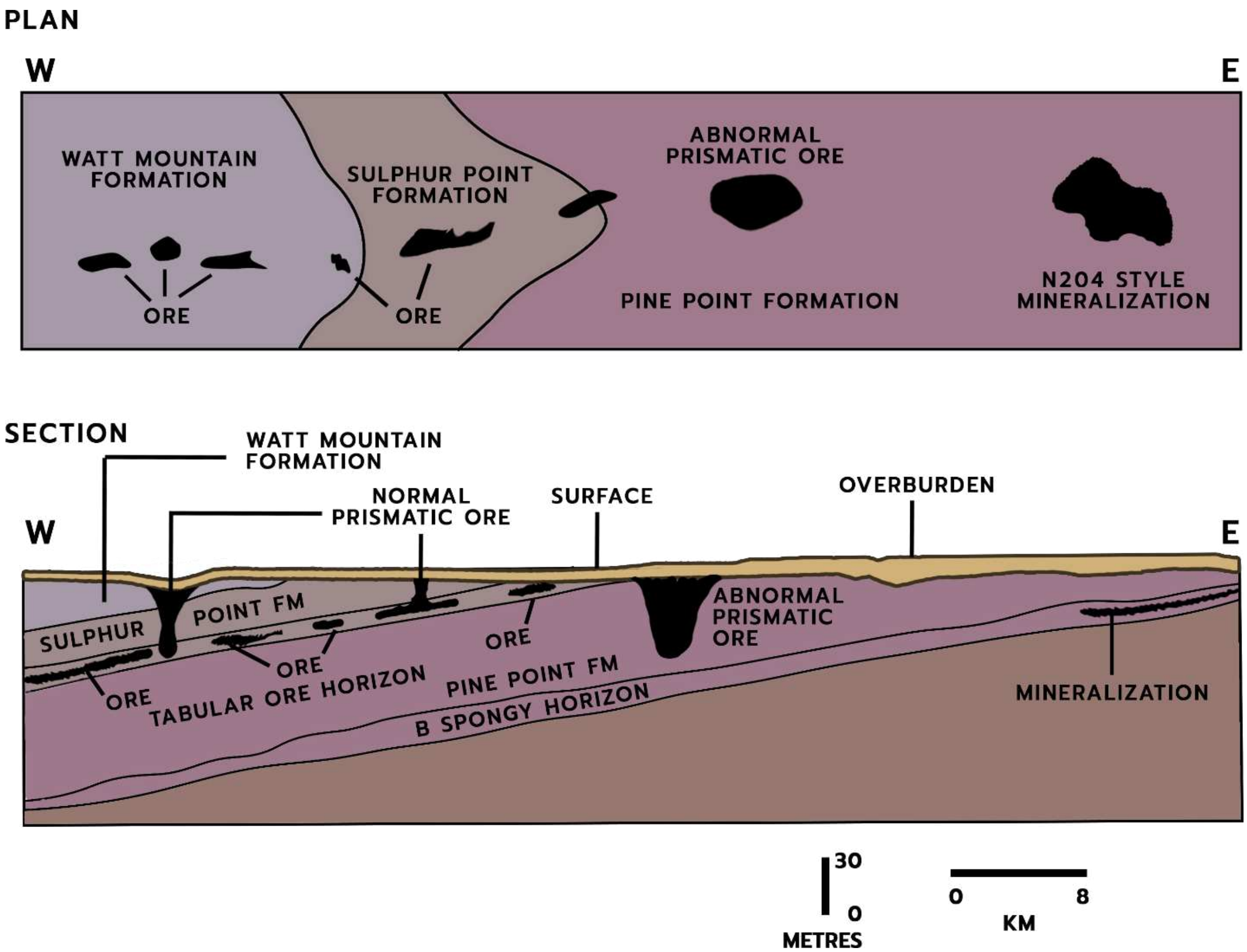


Figure 3: Schematic diagram showing plan (top) and cross-section (bottom) views of the different orebody types (modified after Rhodes et al., 1984).

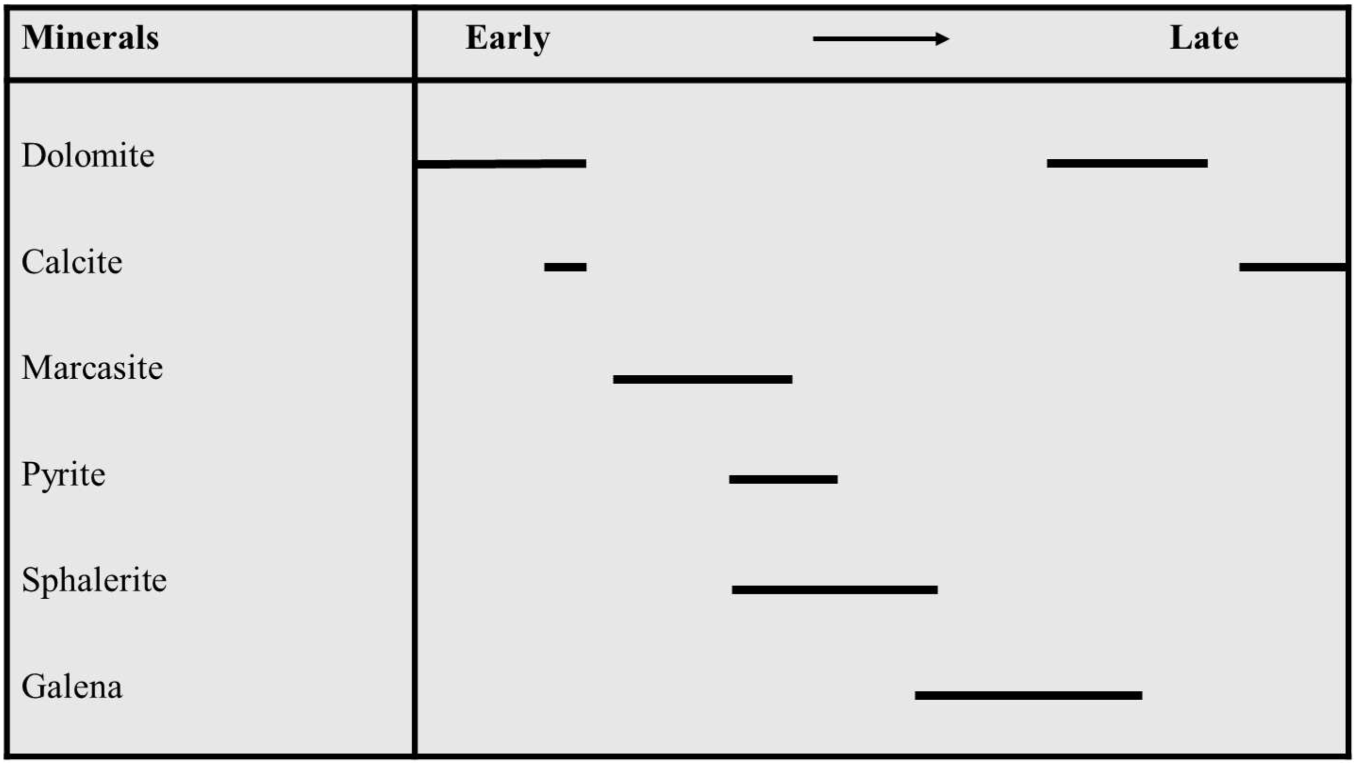


Figure 4: Generalized paragenetic sequence of the Pine Point deposit, showing the relative timing of mineralization.

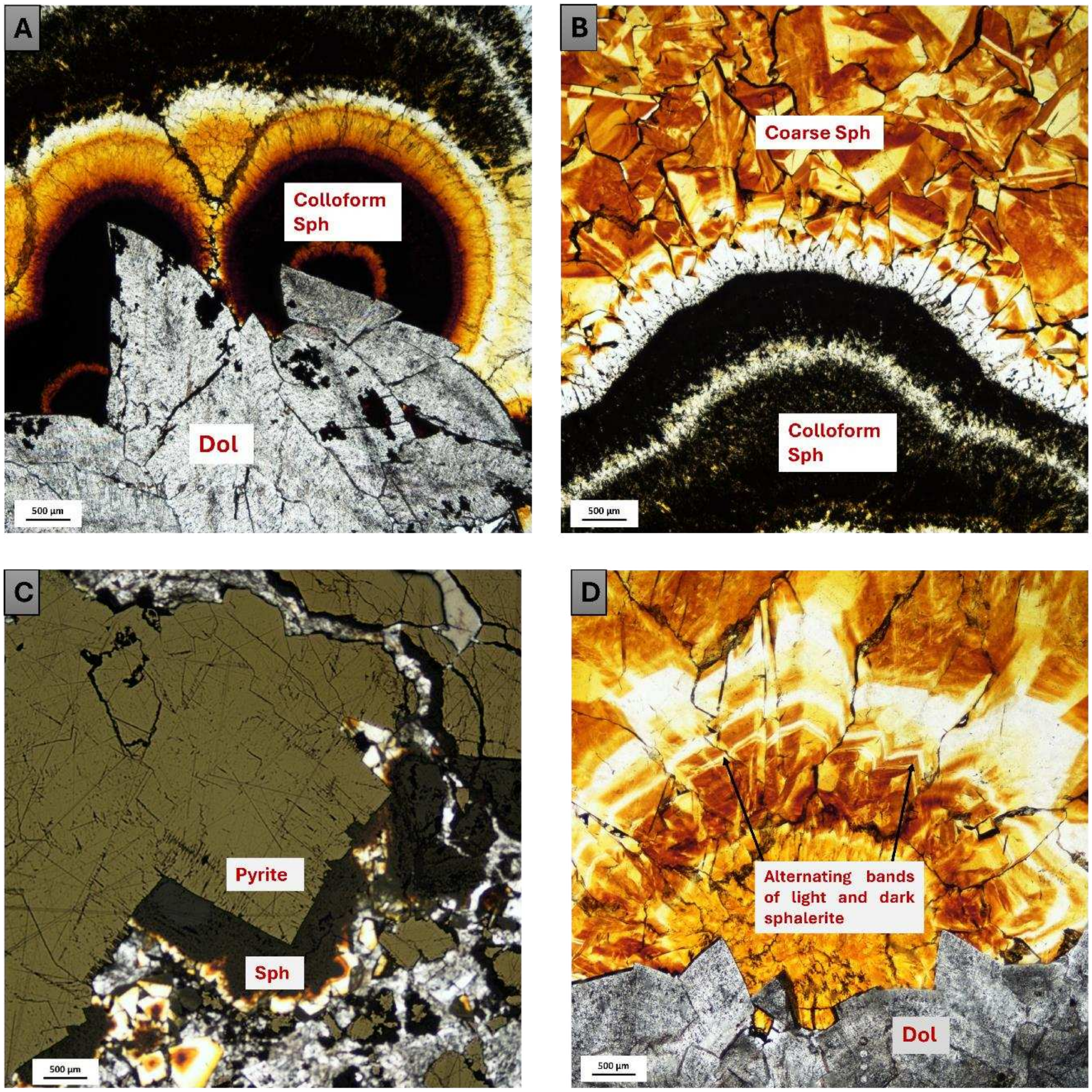
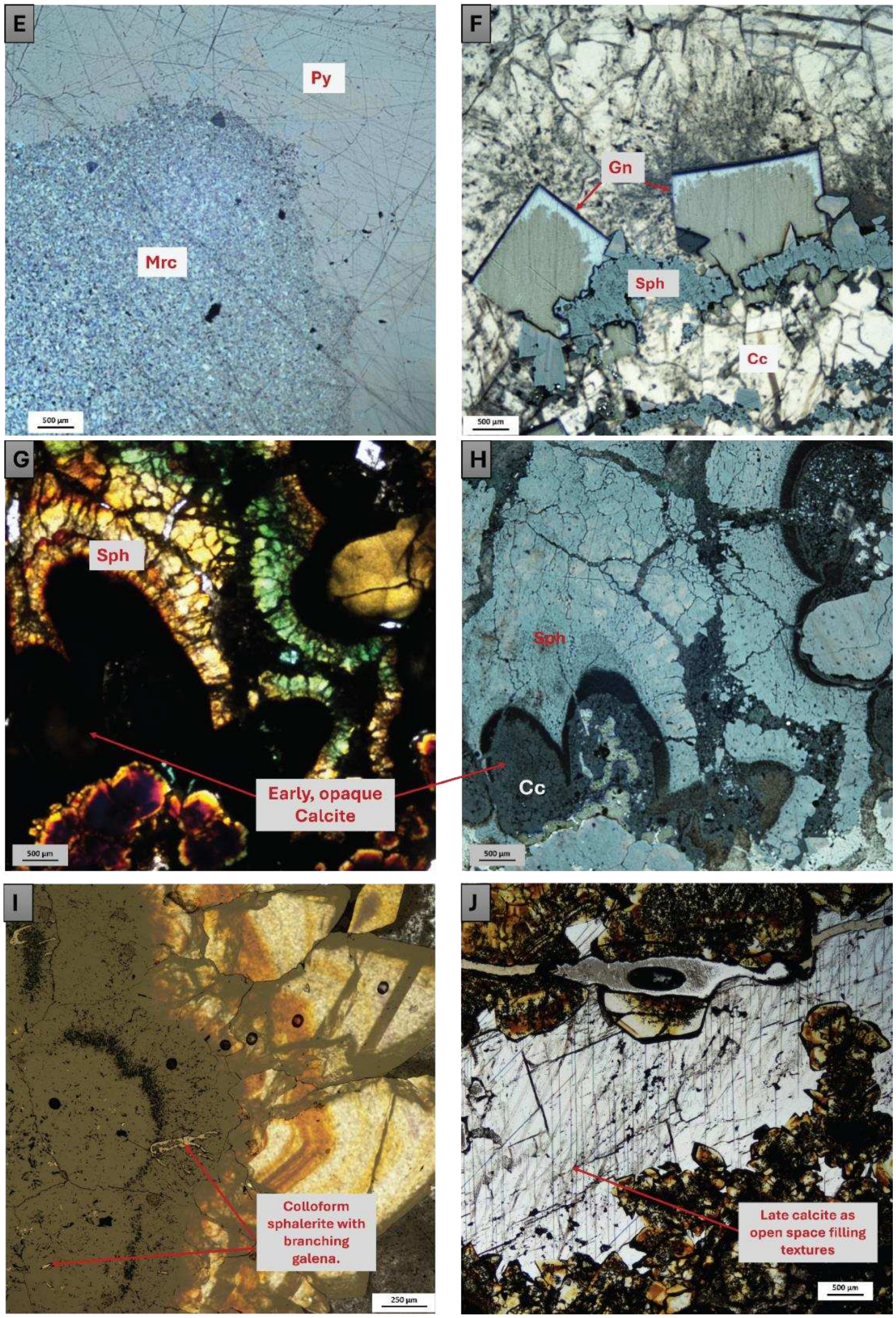


Figure 5: Representative photomicrographs showing the mineral assemblage at Pine Point. (a) Dolomite overgrown by early, colloform sphalerite; (b) The transition from colloform to later formed coarse sphalerite; (c) Early sulphide, pyrite overgrown by colloform and eventually, coarse sphalerite; (d) Alternating pulses of Fe-rich mineralization evidenced by the alternating colour bands; (e) Marcasite overgrown by pyrite; (f) Blocky forms of galena; (g-h) Early, pre-ore calcite overgrown by colloform sphalerite; (i) Early colloform sphalerite coeval with dendritic

galena; (j) Late stage calcite in fractures and open spaces. Abbreviations: Dol=Dolomite; Sph=Sphalerite; Py=Pyrite; Mrc=Marcasite; Gn=Galena; Cc=Calcite.



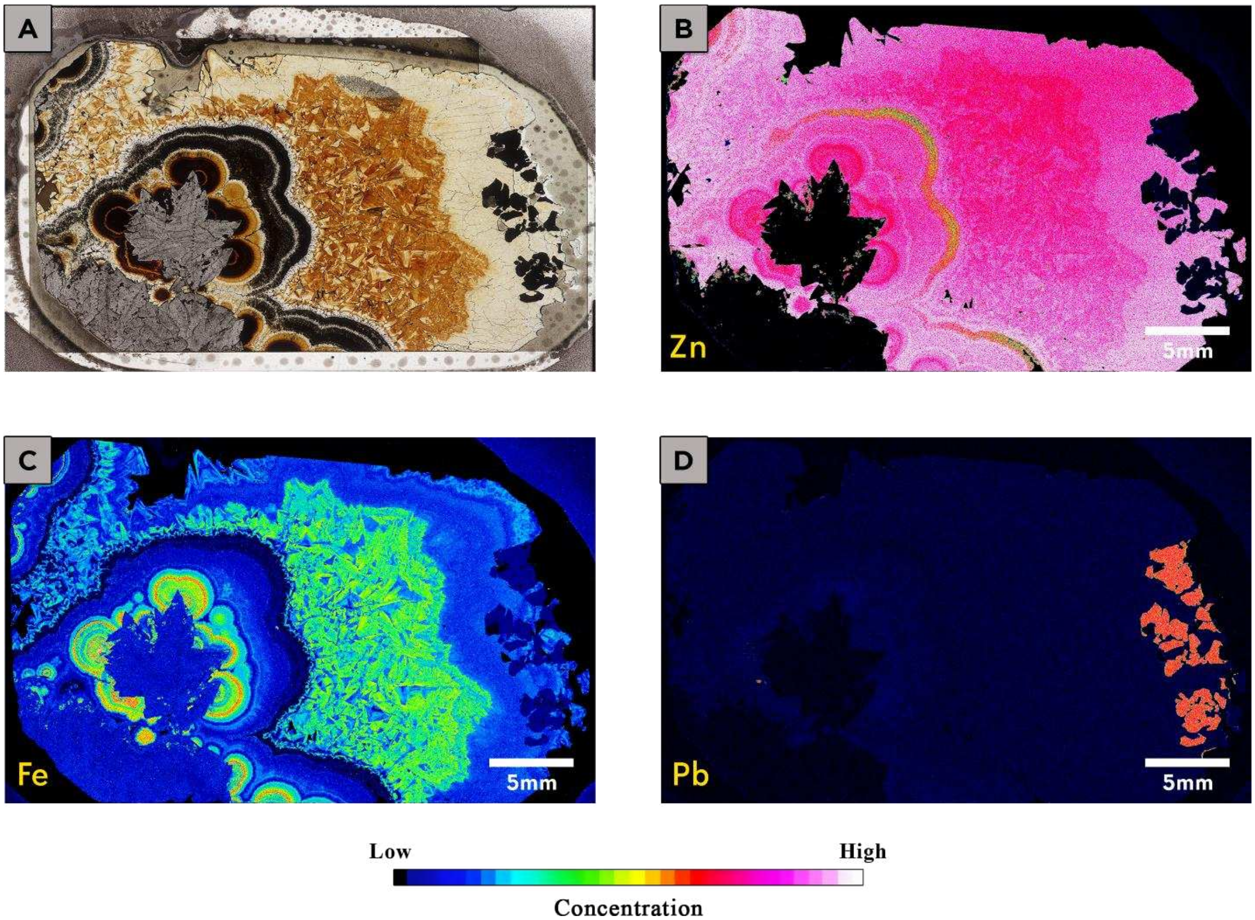
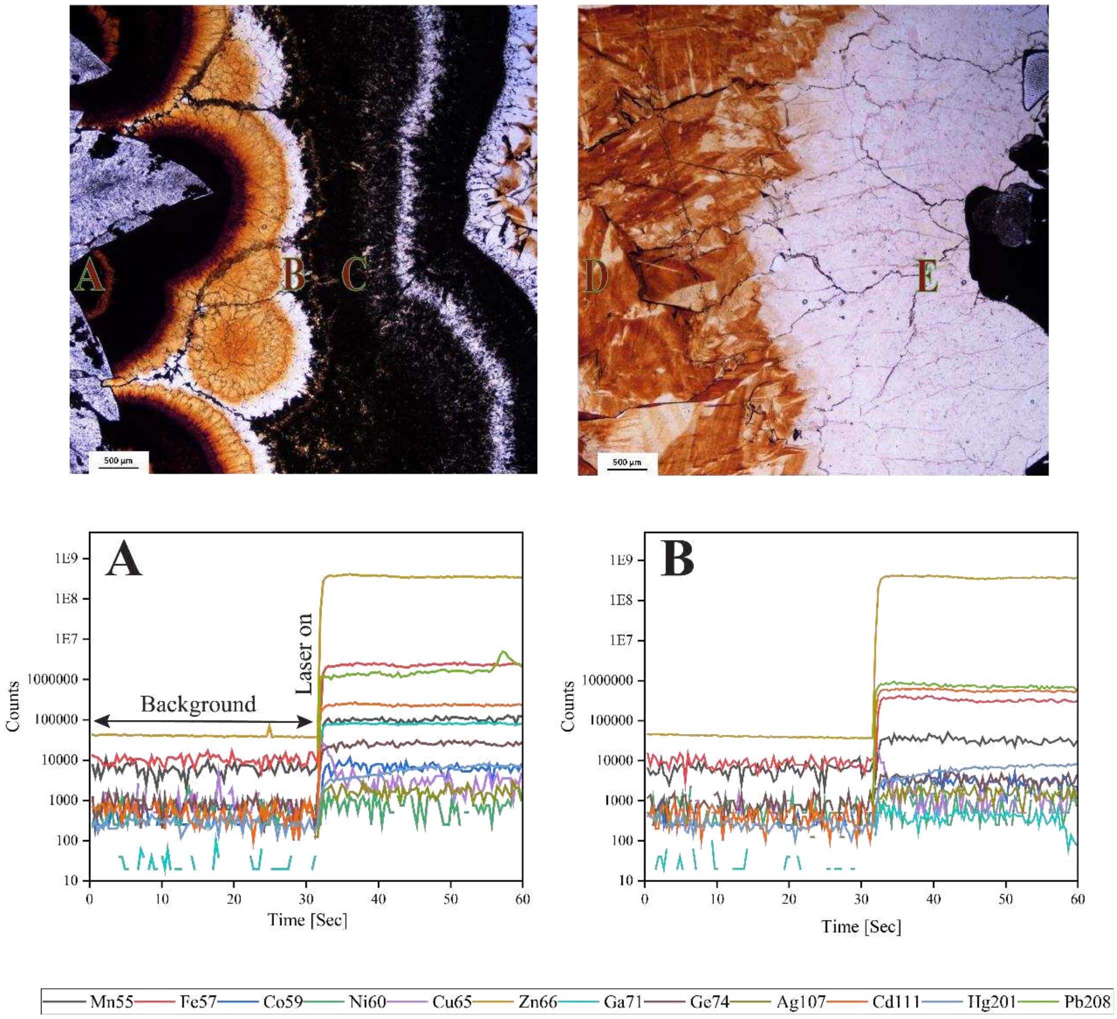


Figure 6: WDS maps of sample O42 showing qualitative concentrations of Zn, Fe, and Pb.


A
B
C
D
E
500 µm
500 µm
A
Laser on
Background
Counts
Time [Sec]
B
Counts
Time [Sec]
1E9
1E8
1E7
1000000
100000
10000
1000
100
10
0
10
20
30
40
50
60
Mn55
Fe57
Co59
Ni60
Cu65
Zn66
Ga71
Ge74
Ag107
Cd111
Hg201
Pb208

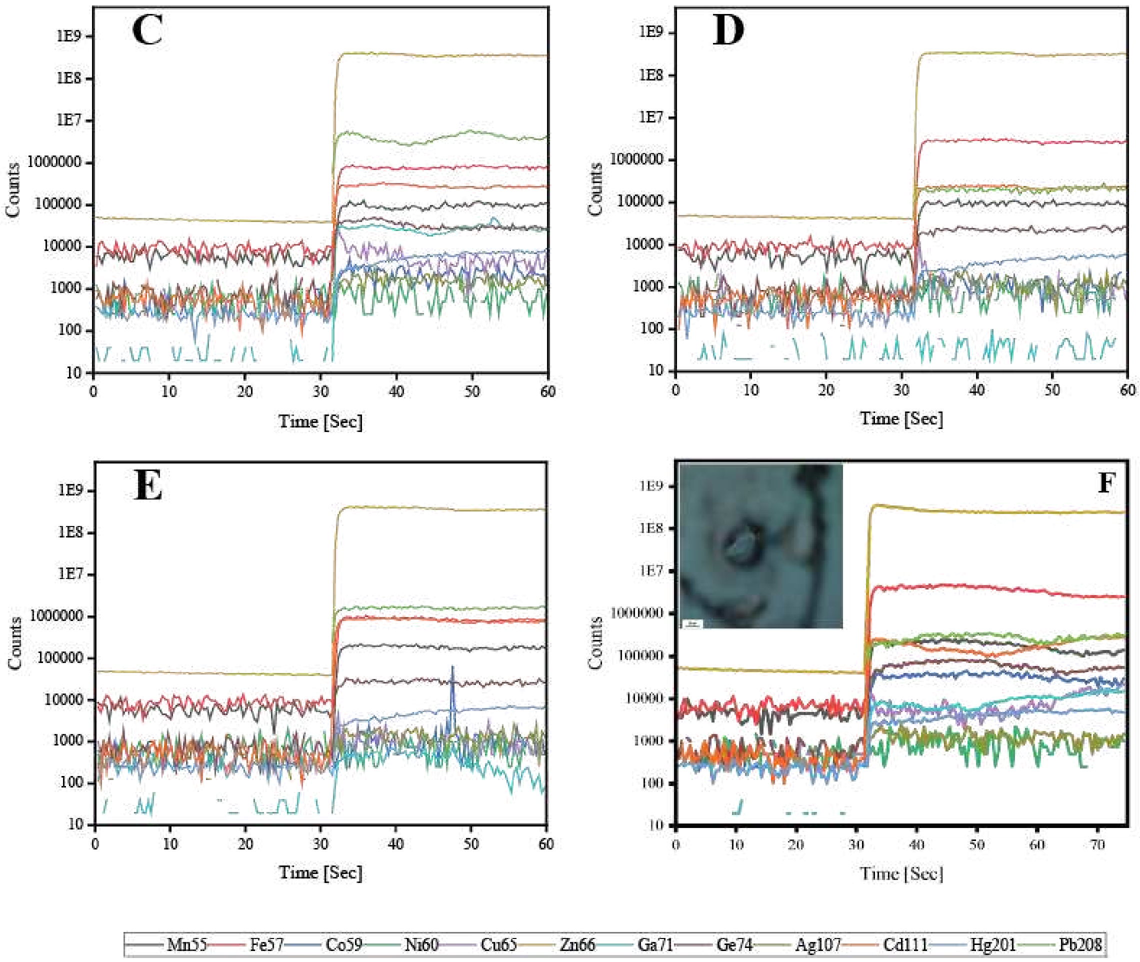


Figure 7 a-e: Representative LA-ICP-MS time-resolved depth profiles of selected elements across the different sphalerite textures in sample O42; (f) Time- resolved signals showing mineral inclusion in sample N204.

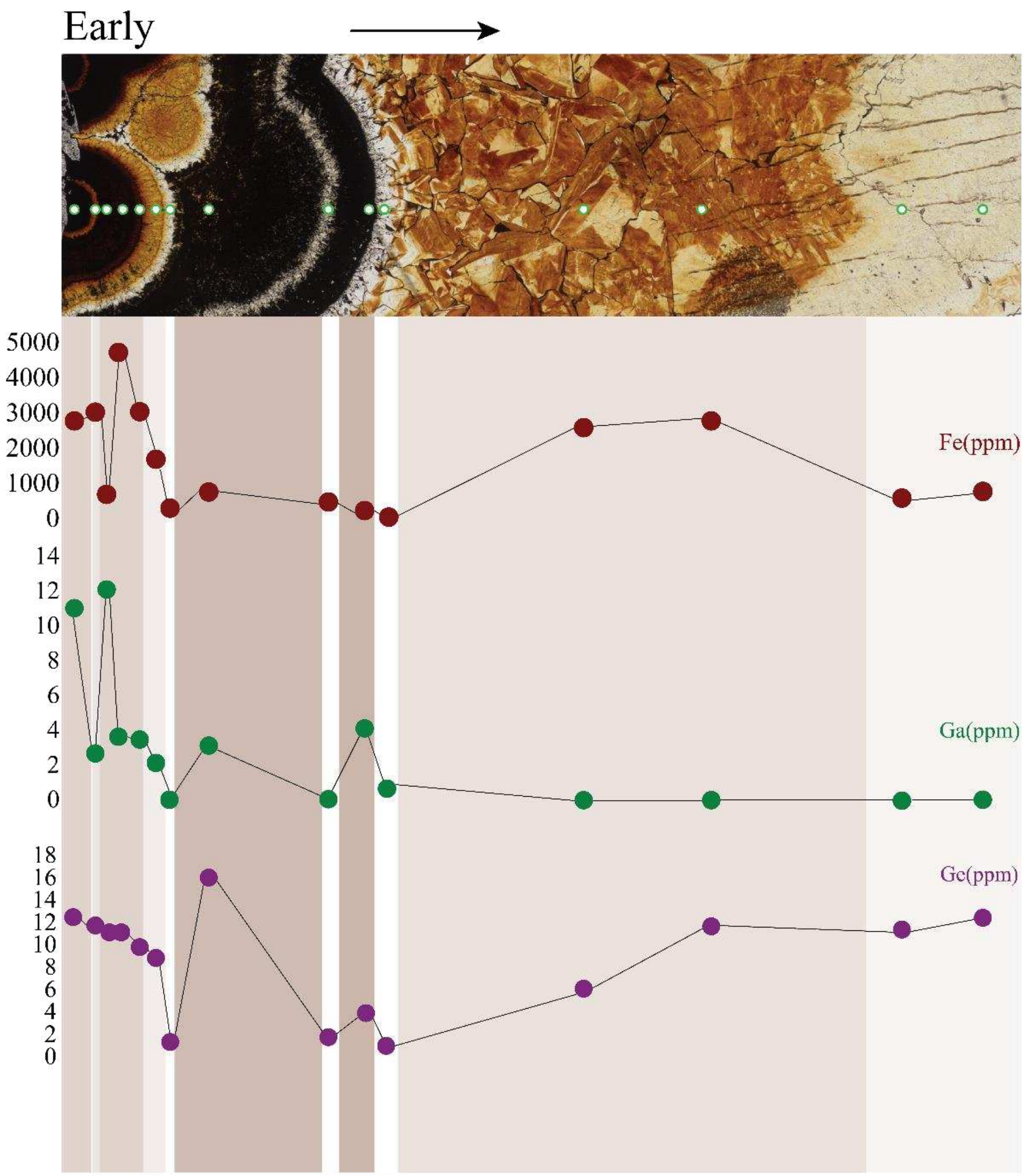


Figure 8: Detailed traverse obtained for sample O42, a prismatic orebody; showing the variation in Fe, Ga and Ge from early to late and across different textures.

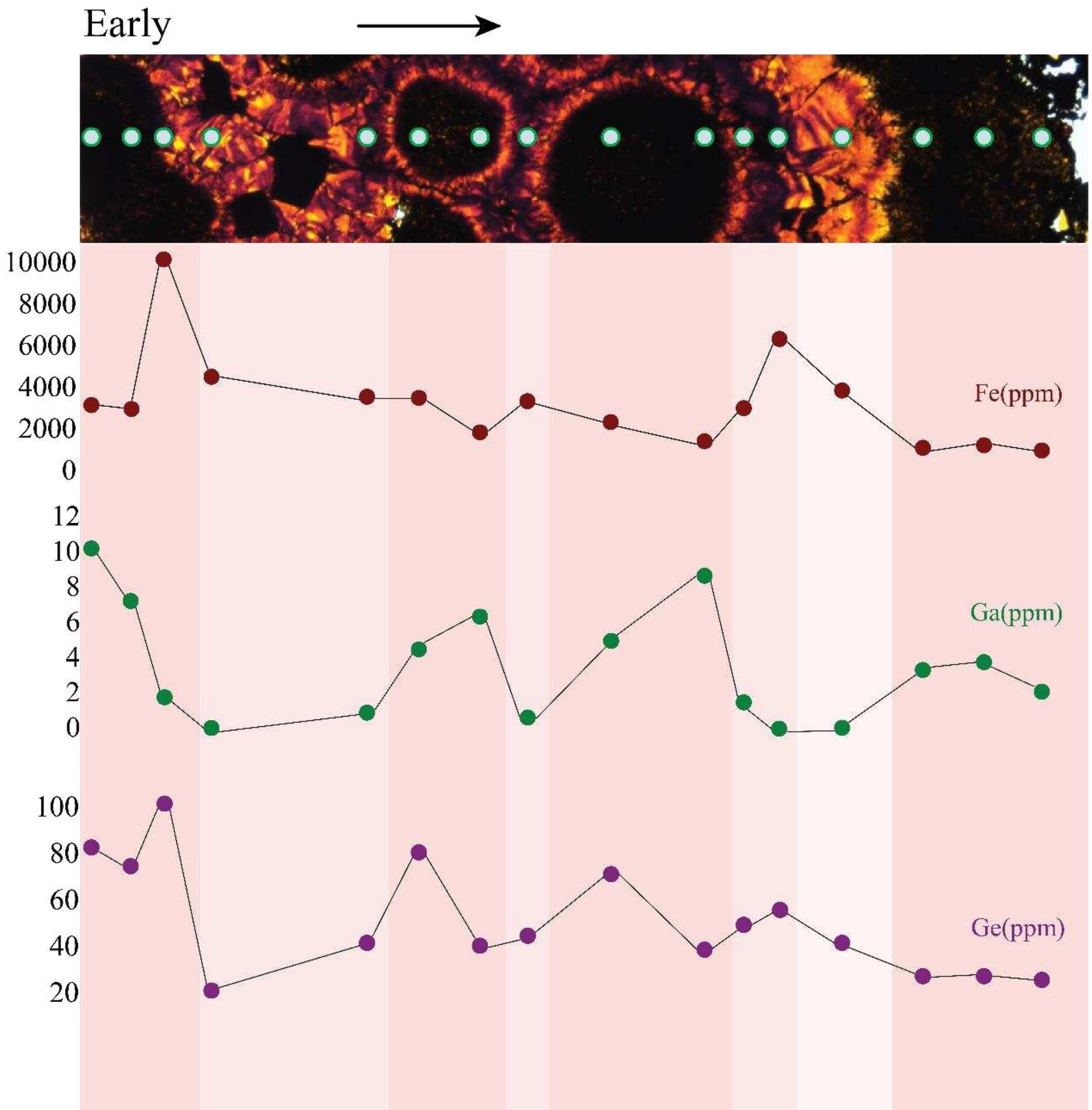


Figure 9: Detailed traverse across the X-15-28 sample, which is an abnormal orebody; showing the variation in Fe, Ga and Ge from early to late and across different textures.

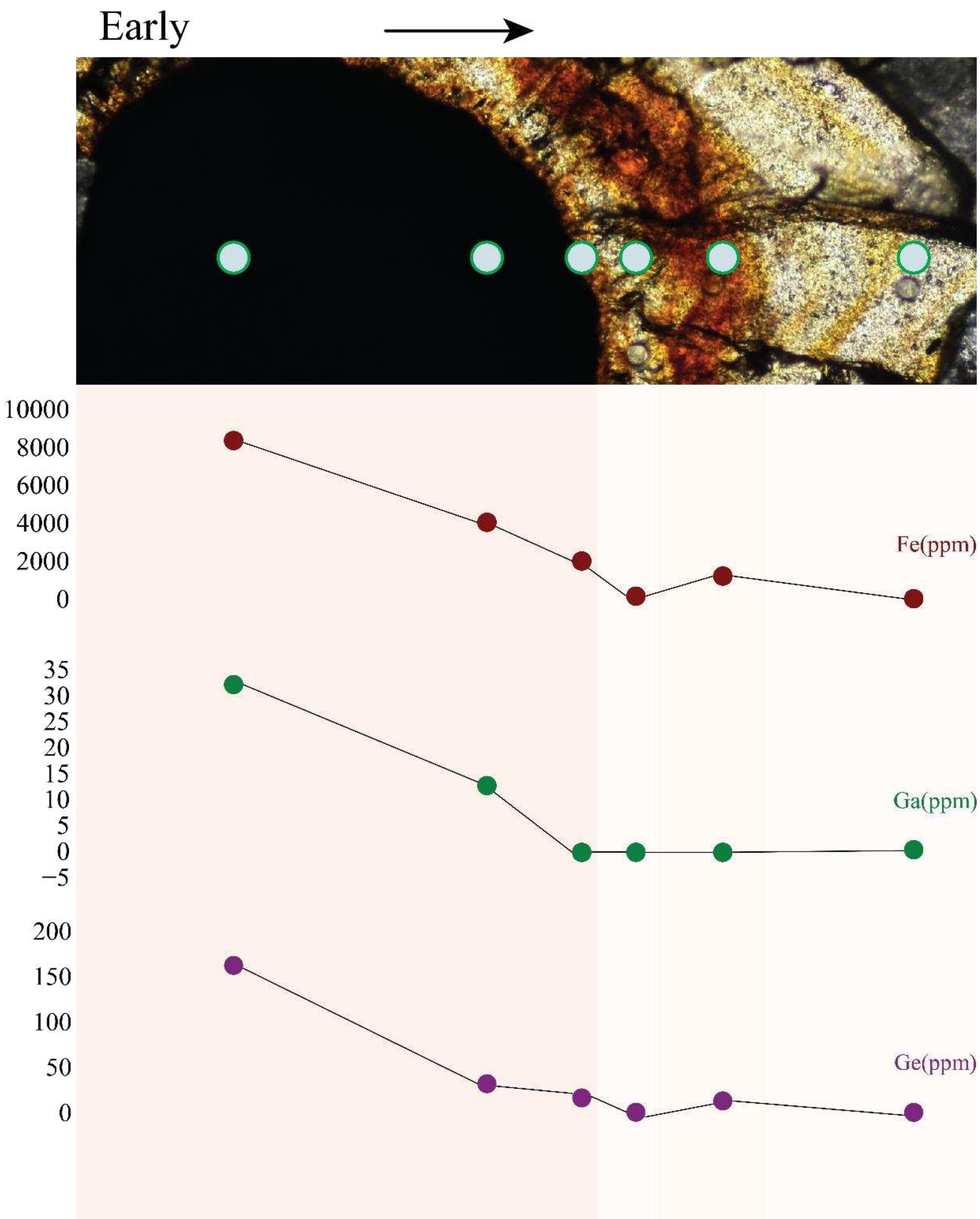
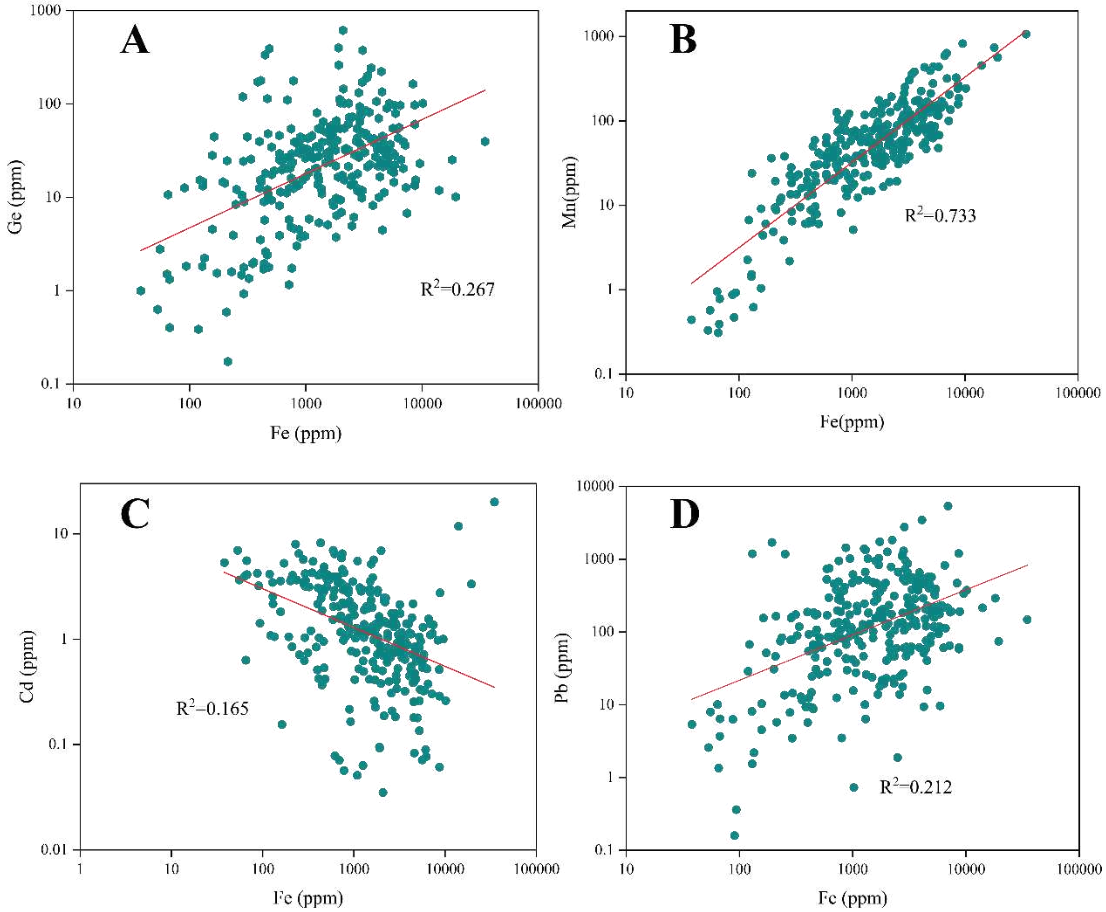


Figure 10: Detailed traverse across the M40-176 sample, which is a tabular orebody; showing the variation in Fe, Ga and Ge from early to late and across different textures.



A
1000
100
10
1
0.1
Ge (ppm)
R^2=0.267
10
100
1000
10000
100000
Fe (ppm)
B
1000
100
10
1
0.1
Mn(ppm)
R^2=0.733
10
100
1000
10000
100000
Fe(ppm)
C
10
1
0.1
0.01
Cd (ppm)
R^2=0.165
1
10
100
1000
10000
100000
Fe (ppm)
D
10000
1000
100
10
1
0.1
Pb (ppm)
R^2=0.212
10
100
1000
10000
100000
Fe (ppm)

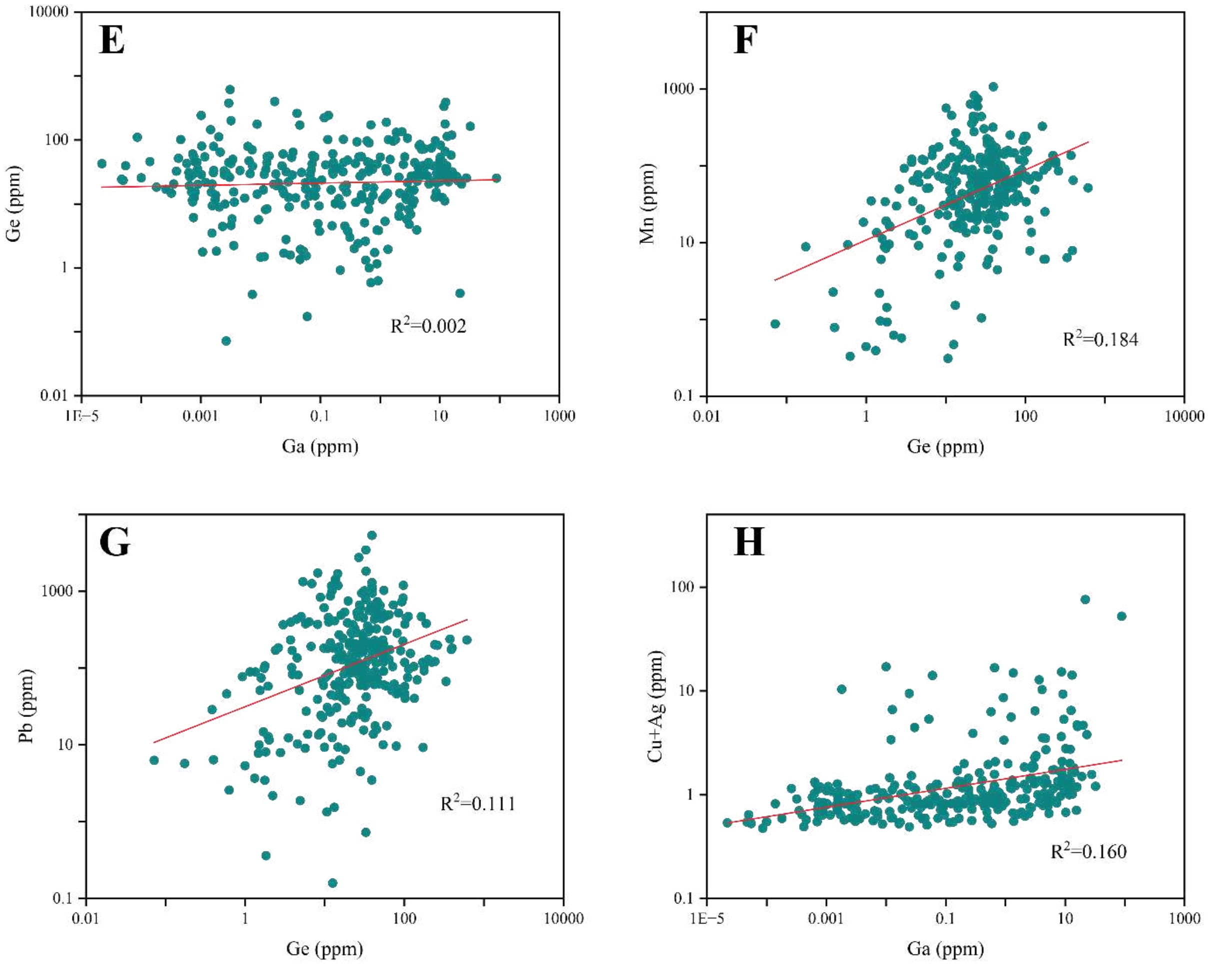


Figure 11: Correlation plots showing suggested coupled substitutions of (a) Fe vs Ge; (b) Fe vs Mn; (c) Fe vs Cd; (d) Fe vs Pb; (e) Ge vs Ga; (f) Ge vs Mn; (g) Ge vs Pb; (h) (Cu+Ag) vs Ga.

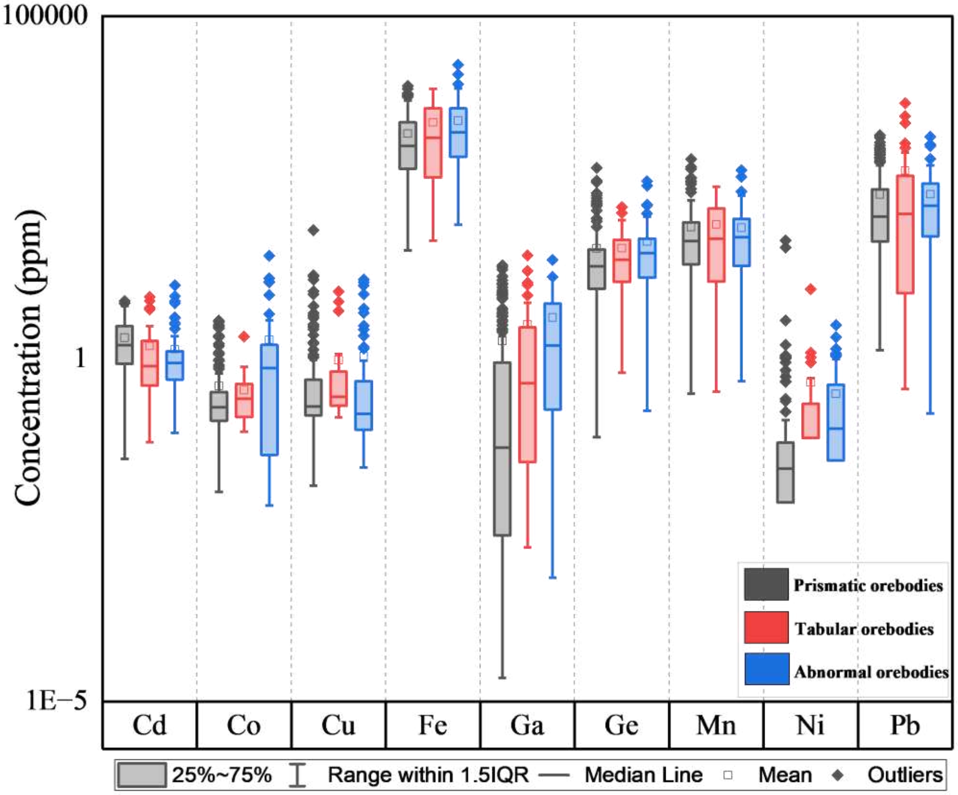


Figure 12: Box plots of trace elements showing variations in sphalerite of the three different deposit types. Mean values are shown as white squares and median values as dark lines.

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Appendix

Appendix A: Summarised EPMA values for sphalerite								
Elements		**Zn WT%**	**Cd WT%**	**Pb WT%**	**Fe WT%**	**Cu WT%**	**S WT%**	**As WT%**
Sample number								
M40.18.PP.135HT12	Mean	63.41	0.02	0.15	3.42	0.00	33.01	0.00
	Max	66.89	0.17	0.33	6.80	0.01	33.35	0.01
	Min	59.90	0.00	0.00	0.46	0.00	32.76	0.00
	STDEV	1.46	0.03	0.10	1.35	0.00	0.14	0.00
N32A-2F	Mean	62.34	0.05	0.03	3.58	0.00	32.50	0.00
	Max	66.32	0.36	0.17	10.39	0.00	32.88	0.01
	Min	54.82	0.00	0.00	0.11	0.00	32.14	0.00
	STDEV	3.81	0.09	0.05	3.47	0.00	0.23	0.00
M40.176.9/12HT12	Mean	62.64	0.06	0.18	3.59	0.00	32.80	0.00
	Max	66.72	0.31	1.34	12.27	0.01	33.46	0.01
	Min	53.51	0.00	0.00	0.09	0.00	31.80	0.00
	STDEV	4.55	0.09	0.33	4.17	0.00	0.38	0.00
W17.44.6HT12	Mean	63.69	0.10	0.05	2.51	0.00	32.92	0.00
	Max	67.00	0.30	0.25	9.26	0.02	33.49	0.01
	Min	56.49	0.00	0.00	0.05	0.00	32.39	0.00
	STDEV	2.62	0.08	0.06	2.42	0.00	0.22	0.00
O42-1	Mean	64.30	0.12	0.06	1.96	0.00	32.74	0.00
	Max	66.65	0.36	0.87	7.99	0.04	33.25	0.00
	Min	58.06	0.00	0.00	0.08	0.00	30.21	0.00
	STDEV	1.73	0.07	0.14	1.58	0.01	0.41	0.00
W17.44.6HT11	Mean	64.03	0.08	0.08	2.57	0.00	32.28	0.00
	Max	67.04	0.28	0.79	9.50	0.00	32.71	0.01
	Min	56.46	0.00	0.00	0.11	0.00	31.65	0.00
	STDEV	2.36	0.06	0.12	2.12	0.00	0.20	0.00
K62C	Mean	65.16	0.11	0.01	1.30	0.00	33.09	0.00
	Max	67.17	0.40	0.05	3.93	0.02	33.56	0.01
	Min	62.46	0.00	0.00	0.04	0.00	32.63	0.00
	STDEV	1.08	0.10	0.01	1.05	0.00	0.14	0.00
K57BFIT12	Mean	65.20	0.10	0.04	1.95	0.00	32.79	0.00

	Max	67.44	0.46	0.43	6.15	0.01	33.39	0.01
	Min	60.84	0.00	0.00	0.07	0.00	32.16	0.00
	STDEV	1.58	0.09	0.07	1.48	0.00	0.24	0.00
W17.46.203	Mean	64.39	0.11	0.15	2.00	0.00	32.30	0.00
	Max	67.29	0.67	0.76	6.64	0.01	32.77	0.00
	Min	59.04	0.00	0.00	0.08	0.00	31.63	0.00
	STDEV	2.23	0.13	0.19	1.92	0.00	0.29	0.00
O28-2	Mean	64.05	0.12	0.05	2.43	0.04	32.66	0.00
	Max	66.80	0.44	0.40	9.20	0.13	33.30	0.00
	Min	56.96	0.01	0.00	0.24	0.02	32.10	0.00
	STDEV	1.67	0.09	0.08	1.45	0.01	0.32	0.00
N38A-4	Mean	62.98	0.10	0.17	2.84	0.00	32.88	0.02
	Max	66.23	0.29	0.51	7.82	0.00	33.29	0.05
	Min	57.15	0.01	0.00	0.14	0.00	32.21	0.01
	STDEV	2.26	0.06	0.13	2.02	0.00	0.18	0.01
N204.ISA.77HT12	Mean	64.46	0.07	0.11	2.29	0.00	33.05	0.00
	Max	67.31	0.20	0.80	10.00	0.01	33.47	0.01
	Min	56.00	0.00	0.00	0.17	0.00	32.58	0.00
	STDEV	2.64	0.06	0.20	2.32	0.00	0.22	0.00
X15.28.02	Mean	61.77	0.02	0.07	4.36	0.00	33.10	0.00
	Max	65.34	0.13	0.74	13.57	0.01	33.52	0.00
	Min	51.77	0.00	0.00	0.92	0.00	32.44	0.00
	STDEV	2.39	0.02	0.11	2.18	0.00	0.21	0.00
X15.393-5HT11	Mean	63.65	0.12	0.07	2.72	0.00	32.28	0.00
	Max	66.46	0.57	0.28	11.07	0.00	32.75	0.00
	Min	54.43	0.00	0.00	0.08	0.00	31.46	0.00
	STDEV	3.13	0.14	0.06	2.93	0.00	0.25	0.00

Appendix B: Summarised LA-ICP-MS values for sphalerite from different samples

	S34_ppm	Mn55_ppm	Fe57_ppm	Co59_ppm	Ni60_ppm	Cu63_ppm	Cu65_ppm	Zn66_ppm	Ga71_ppm	Ge72_ppm	Ge73_ppm	Ge74_ppm	As75_ppm	Ag107_ppm	Cd111_ppm	In115_ppm	Sn118_ppm	Sb121_ppm	Hg201_ppm	Tl205_ppm	Pb208_ppm
Sample number																					
O42	291500.00	58.70	2765.00	0.58	0.17	0.98	0.75	656000.00	11.03	13.48	13.02	12.51	0.03	0.63	1.31	0.00	0.30	0.00	0.19	0.20	183.00
O42	285100.00	128.40	3066.00	0.39	0.09	0.17	0.19	638000.00	2.74	12.75	12.58	11.81	0.02	0.56	1.10	0.00	0.02	0.00	0.16	0.76	872.00
O42	262800.00	47.80	684.00	0.25	0.07	0.81	0.77	606000.00	12.11	11.84	11.23	11.14	0.02	0.59	2.05	0.00	0.47	0.00	0.21	0.41	457.80
O42	276000.00	126.10	4735.00	0.55	0.24	0.27	0.30	598000.00	3.68	12.46	12.09	11.22	0.02	0.54	0.77	0.00	0.08	0.00	0.13	0.14	399.00
O42	261400.00	106.00	3035.00	0.41	0.04	0.48	0.42	581000.00	3.48	10.73	10.26	9.84	0.02	0.58	0.53	0.00	0.05	0.00	0.16	0.69	613.00
O42	277900.00	97.10	1697.00	0.24	0.07	0.29	0.38	633000.00	2.16	9.65	9.57	8.84	0.01	0.54	1.21	0.00	0.01	0.00	0.20	1.34	838.00
O42	287400.00	13.45	323.50	0.20	0.08	0.13	0.22	666000.00	0.05	1.88	1.52	1.35	0.01	0.60	3.13	0.00	0.00	0.00	0.24	0.57	89.00
O42	285300.00	43.30	749.00	0.16	0.09	1.76	1.66	654000.00	3.10	17.51	17.52	16.04	0.01	0.62	1.55	0.00	0.19	0.00	0.25	0.88	491.00
O42	284600.00	19.06	480.00	0.13	0.10	0.21	0.31	658000.00	0.05	2.32	1.87	1.78	0.01	0.57	4.00	0.00	0.01	0.00	0.27	0.79	107.30
O42	317400.00	12.25	236.60	0.09	0.26	10.11	9.65	665000.00	4.12	4.51	4.22	3.92	0.01	0.64	3.64	0.00	0.13	0.00	0.57	0.26	83.00
O42	281000.00	0.44	38.10	0.04	0.05	15.55	15.92	655000.00	0.66	1.44	0.96	1.00	0.02	0.81	5.32	0.00	0.00	0.00	0.25	0.02	5.36
O42	292900.00	51.40	2594.00	0.03	0.00	0.09	0.25	654000.00	0.00	7.17	6.91	6.12	0.01	0.60	1.76	0.00	0.00	0.00	0.26	0.00	13.80
O42	276800.00	43.90	2785.00	0.07	0.06	0.10	0.09	621000.00	0.00	13.09	13.09	11.69	0.01	0.56	1.38	0.00	0.00	0.00	0.27	0.04	26.84
O42	281000.00	35.20	608.10	0.06	0.10	0.11	0.18	656000.00	0.00	12.47	12.30	11.39	0.01	0.57	2.91	0.00	0.00	0.00	0.35	0.12	85.20
O42	277500.00	79.10	782.60	0.06	0.06	0.09	0.19	652000.00	0.04	13.37	12.52	12.44	0.01	0.57	4.28	0.00	0.00	0.00	0.37	0.28	189.60
K62C	265600.00	0.57	55.50	0.08	-0.04	0.05	0.17	671000.00	0.03	3.34	2.81	2.78	0.01	1.36	3.64	0.00	0.01	0.00	1.45	0.06	7.95
K62C	266600.00	0.95	64.00	0.08	0.04	0.07	0.21	670000.00	0.01	2.04	1.65	1.50	0.01	1.25	3.98	0.00	0.01	0.00	1.99	0.06	10.05
K62C	266500.00	12.96	408.70	0.15	-0.01	0.04	0.15	664000.00	0.00	4.16	3.57	3.50	0.01	1.12	4.03	0.00	0.01	0.00	2.92	0.03	12.38
K62C	265200.00	52.70	1735.00	0.30	0.03	0.07	0.17	638000.00	0.00	63.42	61.00	60.20	0.01	1.00	0.53	0.00	0.00	0.00	3.19	0.05	17.77
K62C	264700.00	8.39	292.30	0.09	-0.01	0.06	0.18	655000.00	0.00	2.35	2.04	1.77	0.01	1.02	4.19	0.00	0.00	0.00	5.25	0.01	3.46
K62C	266100.00	12.39	1131.00	0.35	-0.03	0.05	0.16	644000.00	0.00	47.90	46.90	45.70	0.01	1.02	0.52	0.00	0.00	0.00	5.49	0.02	74.80
K62C	266000.00	30.11	721.10	0.24	-0.01	0.06	0.18	650000.00	0.00	13.11	12.99	12.45	0.01	0.97	1.67	0.00	0.00	0.00	4.93	0.03	12.45
K62C	274900.00	66.90	1811.00	0.25	0.00	0.03	0.16	652000.00	0.00	23.04	22.40	21.63	0.01	0.92	0.72	0.00	0.00	0.00	2.86	0.01	21.68
K62C	266900.00	36.23	2759.00	0.43	-0.04	0.06	0.13	625000.00	0.00	45.14	43.90	42.93	0.01	0.82	0.74	0.00	0.01	0.00	1.55	0.03	17.69
K62C	276300.00	32.21	1025.00	0.22	0.01	0.10	0.19	664000.00	0.00	33.20	33.00	31.30	0.01	0.86	1.72	0.00	0.00	0.00	1.38	0.02	15.82
K62C	271900.00	6.47	420.80	0.32	0.00	0.05	0.19	662000.00	0.00	16.02	15.11	14.85	0.01	0.73	1.04	0.00	0.00	0.00	0.86	0.02	9.50
K62C	267900.00	30.63	1260.00	0.26	-0.05	0.06	0.07	628000.00	0.00	15.88	16.07	14.92	0.02	0.55	1.84	0.00	0.00	0.00	0.18	0.01	6.16
K62C	290100.00	111.20	4387.00	0.23	0.01	0.05	0.11	648000.00	0.00	11.58	11.38	10.27	0.01	0.50	0.57	0.00	0.00	0.00	0.14	0.00	9.66
K62C	279300.00	9.00	155.50	0.33	-0.04	0.81	0.78	654000.00	0.16	5.23	4.88	4.49	0.01	0.92	3.52	0.00	0.01	0.00	0.19	0.03	10.19
K62C	279900.00	0.86	86.30	0.15	-0.02	0.12	0.22	655000.00	0.00	0.52	0.00	0.07	0.02	0.54	4.16	0.00	0.00	0.00	0.18	0.01	6.20
M40-176	260900.00	241.00	8630.00	0.76	1.23	0.20	0.36	600000.00	8.40	103.50	104.60	97.30	0.23	0.61	0.52	0.00	0.10	0.00	0.06	0.74	1200.00
M40-176	210000.00	317.00	6940.00	0.26	0.89	0.41	0.65	635000.00	12.00	41.20	41.10	39.20	0.22	1.35	1.39	0.00	0.40	0.00	0.43	3.76	5360.00
M40-176	190900.00	118.40	2860.00	0.21	0.52	0.32	0.38	615000.00	19.08	28.02	26.00	26.80	0.09	1.12	1.42	0.00	0.92	0.00	0.67	1.72	2760.00
M40-176	193200.00	11.19	229.90	0.09	-0.01	0.11	0.35	644000.00	0.03	2.08	1.75	1.59	0.02	0.78	7.97	0.00	0.00	0.00	0.83	0.23	74.60
M40-176	201300.00	56.30	1363.00	0.26	0.10	0.13	0.29	655000.00	0.01	13.64	13.40	12.35	0.01	0.83	0.78	0.00	0.00	0.00	0.73	0.26	1016.00
M40-176	204900.00	0.33	53.30	0.09	0.09	6.34	6.78	667000.00	0.92	1.06	0.61	0.63	0.02	1.83	6.95	0.00	0.00	0.00	0.71	0.02	2.58
M40-176	205600.00	0.39	67.10	0.14	0.05	5.04	4.98	664000.00	0.58	1.89	1.28	1.32	0.03	1.31	5.55	0.00	0.00	0.00	0.63	0.01	3.68
M40-176	208700.00	62.30	1263.00	0.24	-0.06	0.10	0.24	653000.00	0.00	14.80	14.62	13.68	0.01	0.81	0.62	0.00	0.01	0.00	0.54	0.04	1392.00
M40-176	211900.00	6.02	172.40	0.12	-0.09	0.08	0.28	657000.00	0.06	2.02	1.65	1.54	0.01	0.85	5.25	0.00	0.00	0.00	0.60	0.15	51.40
M40-176	215700.00	90.10	2058.00	0.27	-0.09	0.12	0.16	626000.00	0.06	18.75	17.80	17.21	0.02	0.75	0.50	0.00	0.00	0.00	0.56	0.08	130.30
M40-176	221200.00	225.60	4091.00	0.24	0.27	0.21	0.46	612000.00	12.86	34.48	34.90	32.89	0.07	1.18	1.14	0.00	0.41	0.00	0.38	2.22	3460.00
M40-176	218900.00	325.40	8357.00	0.52	1.05	0.13	0.21	538000.00	32.30	171.00	169.40	163.40	0.18	1.00	0.98	0.00	0.28	0.00	0.24	1.08	469.20
M40-18-PP	235500.00	14.60	442.00	0.32	-0.04	0.14	0.18	656000.00	0.01	6.41	5.53	5.69	0.01	0.81	2.99	0.00	0.00	0.00	0.66	0.01	9.05
M40-18-PP	243100.00	32.10	1617.00	0.60	0.15	0.23	0.29	649000.00	0.11	24.99	23.70	23.69	0.02	0.73	0.61	0.00	0.00	0.00	0.57	0.03	333.00
M40-18-PP	249100.00	156.90	4065.00	0.62	0.26	1.20	1.17	623000.00	2.87	111.30	109.20	106.20	0.03	0.76	0.45	0.00	0.00	0.00	0.40	0.19	451.50
M40-18-PP	252000.00	213.60	2907.00	0.43	0.22	0.54	0.51	631000.00	0.41	97.82	97.00	93.30	0.02	0.87	0.45	0.00	0.00	0.00	0.47	0.25	732.00
M40-18-PP	252400.00	250.30	2932.00	0.42	0.23	0.11	0.23	620000.00	0.00	67.99	67.90	64.45	0.02	0.61	0.41	0.00	0.00	0.00	0.43	0.19	495.20
M40-18-PP	310600.00	88.40	4990.00	2.11	10.40	0.72	0.93	635000.00	0.65	94.80	92.30	89.80	0.40	0.64	0.18	0.00	0.01	0.00	0.24	0.26	169.00
M40-18-PP	275400.00	13.38	448.20	0.26	-0.06	0.05	0.28	669000.00	0.24	19.34	18.90	18.06	0.01	0.73	1.87	0.00	0.01	0.00	0.51	0.02	8.71
M40-18-PP	271600.00	16.70	400.80	0.32	-0.09	0.08	0.14	656000.00	0.48	13.53	13.50	12.50	0.01	0.70	2.15	0.00	0.01	0.00	0.48	0.01	5.69

N32A-2F	234300.00	36.50	1669.00	0.11	0.03	0.06	0.23	548000.00	0.12	43.10	42.60	40.80	0.01	0.60	0.26	0.00	0.00	0.00	0.35	0.09	13.82
N32A-2F	293600.00	170.70	6235.00	0.43	0.03	0.10	0.24	629000.00	0.69	45.49	44.90	43.03	0.01	0.66	0.08	0.00	0.01	0.00	0.23	0.17	216.30
N32A-2F	320300.00	157.80	8690.00	0.09	-0.01	0.06	0.16	650000.00	0.02	64.30	63.80	60.40	0.01	0.71	0.06	0.00	0.01	0.00	0.20	0.13	58.20
N32A-2F	277200.00	1.04	157.00	0.14	-0.02	0.10	0.22	660000.00	0.07	30.20	29.80	28.10	0.01	0.71	1.82	0.00	0.00	0.00	0.46	0.02	4.51
N32A-2F	290200.00	55.20	2624.00	0.14	0.08	0.05	0.20	657000.00	0.00	30.65	29.20	28.50	0.01	0.75	0.21	0.00	0.00	0.00	0.38	0.02	24.57
N32A-2F	277100.00	142.80	6114.00	0.50	0.05	0.10	0.26	568000.00	1.01	57.25	56.30	53.80	0.03	0.59	0.09	0.00	0.01	0.00	0.21	0.12	232.00
N32A-2F	271300.00	140.90	5623.00	0.48	0.13	0.10	0.19	564000.00	0.49	57.86	58.10	54.70	0.02	0.64	0.07	0.00	0.00	0.00	0.20	0.12	208.40
N32A-2F	256400.00	21.75	1281.00	0.15	0.02	0.11	0.15	584000.00	0.44	56.80	56.20	54.30	0.01	0.62	0.49	0.00	0.00	0.00	0.40	0.07	10.04
N32A-2F	282400.00	0.92	93.60	0.28	-0.11	8.60	9.60	663000.00	0.00	2.50	2.15	1.83	0.01	0.77	1.42	0.00	0.00	0.00	0.42	0.00	0.36
N32A-2F	274100.00	1.52	129.60	0.12	0.00	0.17	0.24	646000.00	0.07	13.94	13.56	13.19	0.01	0.73	2.54	0.00	0.00	0.00	0.39	0.00	1.54
N32A-2F	278900.00	8.18	799.00	0.19	0.21	0.04	0.18	654000.00	0.01	41.49	41.80	39.01	0.01	0.73	1.03	0.00	0.00	0.00	0.39	0.02	3.49
N32A-2F	287500.00	74.60	4530.00	0.33	0.07	0.31	0.41	624000.00	2.31	139.80	138.60	134.20	0.02	0.73	0.18	0.00	0.04	0.00	0.24	0.38	40.50
N32A-2F	272100.00	89.60	4190.00	0.49	0.03	0.65	0.68	601000.00	2.88	23.70	22.80	21.56	0.02	0.71	0.52	0.00	0.05	0.00	0.23	0.18	290.00
N32A-2F	281200.00	37.50	1594.00	0.35	0.09	0.78	0.93	650000.00	6.17	28.80	28.40	27.45	0.02	0.80	1.32	0.00	0.06	0.00	0.36	0.19	155.50
N32A-2F	276900.00	20.80	691.00	0.26	0.18	0.87	0.96	650000.00	6.57	21.41	22.30	20.48	0.02	0.78	1.58	0.00	0.10	0.00	0.42	0.19	130.60
N204-ISA	268200.00	738.00	18140.00	45.10	34.20	-465.00	51.70	667000.00	88.70	26.26	25.90	25.22	0.16	0.63	-61.70	0.00	5.00	0.06	0.48	0.25	290.60
N204-ISA	269400.00	1065.00	34800.00	43.20	10.20	148.00	13.60	636000.00	13.06	41.50	41.50	39.60	0.08	0.63	20.00	0.00	0.17	0.00	0.39	0.47	147.60
N204-ISA	254900.00	453.00	14040.00	31.75	3.10	3.19	1.41	633000.00	1.10	12.57	12.67	11.82	0.06	0.58	11.80	0.00	0.03	0.00	0.39	0.14	214.70
N204-ISA	272400.00	563.00	19450.00	14.98	2.07	0.79	0.75	653000.00	0.18	11.28	10.57	10.08	0.04	0.60	3.34	0.00	0.00	0.00	0.33	0.04	74.40
N204-ISA	247000.00	186.00	8820.00	13.15	1.14	0.55	0.42	608000.00	0.00	16.10	15.90	15.30	0.03	0.59	2.75	0.00	0.00	0.00	0.29	0.05	190.00
N204-ISA	256600.00	86.00	5410.00	13.27	0.72	0.84	0.73	634000.00	0.00	20.30	20.20	19.30	0.02	0.60	0.96	0.00	0.00	0.00	0.32	0.02	390.00
N204-ISA	238700.00	73.90	4572.00	8.53	0.51	0.40	0.46	586000.00	0.00	5.02	4.92	4.45	0.01	0.57	0.47	0.00	0.00	0.00	0.27	0.01	434.70
N204-ISA	264900.00	8.76	432.40	1.94	-0.12	9.77	8.76	670000.00	0.02	2.21	1.81	1.70	0.02	0.69	8.20	0.00	0.00	0.00	0.41	0.08	14.79
N204-ISA	296700.00	99.10	3290.00	1.92	1.40	2.76	3.05	668000.00	0.29	52.15	52.00	48.27	0.01	0.85	1.01	0.00	0.00	0.00	0.38	0.04	62.40
N204-ISA	251100.00	24.39	877.00	0.75	0.61	4.52	4.69	597000.00	1.25	38.55	38.80	35.83	0.01	0.90	1.34	0.00	0.01	0.00	0.42	0.03	67.00
N204-ISA	355500.00	32.70	780.00	0.77	0.75	13.61	14.34	633000.00	8.68	16.50	14.99	15.14	0.02	0.90	2.15	0.00	0.05	0.01	2.76	0.37	272.00
N204-ISA	249500.00	15.72	872.00	0.95	0.77	5.24	5.42	560000.00	3.14	14.77	13.92	13.45	0.01	1.01	1.41	0.00	0.02	0.00	0.34	0.06	43.50
N204-ISA	287400.00	16.52	995.00	0.90	0.58	11.53	11.74	655000.00	3.67	18.15	16.51	16.44	0.01	1.04	2.02	0.00	0.03	0.00	0.40	0.05	41.90
N204-ISA	259800.00	67.20	1986.00	3.15	0.42	0.10	0.09	598000.00	0.00	10.62	9.75	9.36	0.01	0.81	0.38	0.00	0.00	0.00	0.27	0.04	23.00
N204-ISA	264200.00	76.40	1065.00	1.85	0.15	0.15	0.21	624000.00	0.00	6.87	6.34	5.85	0.01	0.80	1.27	0.00	0.00	0.00	0.33	0.08	27.50
N204-ISA	280800.00	8.80	213.30	1.42	1.16	12.44	13.13	666000.00	0.06	0.67	0.05	0.17	0.01	0.96	4.05	0.00	0.01	0.00	0.47	0.00	5.73
N204-ISA	284000.00	83.60	1840.00	3.15	0.65	0.98	0.94	670000.00	2.08	7.67	6.49	6.38	0.01	0.91	0.91	0.00	0.02	0.00	0.35	0.38	401.00
N204-ISA	278600.00	50.70	716.00	2.64	1.09	0.89	0.89	666000.00	2.79	6.72	6.04	5.74	0.03	0.91	1.17	0.00	0.02	0.00	0.39	0.48	378.00
N204-ISA	291200.00	277.00	5210.00	4.49	1.19	2.59	2.65	658000.00	4.18	25.10	24.40	22.70	0.01	0.88	1.05	0.00	0.01	0.00	0.27	0.04	366.00
N204-ISA	269700.00	117.40	1299.00	2.51	0.37	0.38	0.45	649000.00	1.21	6.24	5.71	5.33	0.01	0.83	1.38	0.00	0.01	0.00	0.31	0.72	1337.00
N204-ISA	270400.00	175.20	1734.00	2.88	0.47	0.89	0.94	652000.00	3.00	9.38	8.70	8.18	0.02	0.92	1.39	0.00	0.00	0.00	0.29	0.63	1735.00
N204-ISA	280900.00	162.50	2020.00	3.65	0.99	1.48	1.49	673000.00	3.25	8.12	7.77	6.89	0.03	0.89	1.45	0.00	0.03	0.00	0.33	0.71	1260.00
X15-393	266300.00	33.80	1990.00	0.03	0.04	0.08	0.22	637000.00	0.24	109.50	108.10	102.30	0.02	0.90	6.91	0.00	0.01	0.00	0.35	0.09	40.40
X15-393	271400.00	19.93	616.00	0.05	0.24	0.42	0.49	665000.00	14.36	26.36	25.90	24.30	0.04	1.16	6.44	0.00	0.39	0.00	0.33	0.29	295.00
X15-393	269600.00	54.20	590.00	0.05	0.14	0.13	0.20	660000.00	27.78	27.68	26.71	25.43	0.05	1.37	3.92	0.00	0.96	0.00	0.32	1.75	528.00
X15-393	257700.00	22.80	769.00	0.02	0.05	0.02	0.08	637000.00	9.05	25.92	24.80	23.75	0.02	1.44	0.97	0.00	0.13	0.00	0.27	0.26	311.00
X15-393	257900.00	30.67	2574.00	0.01	0.03	0.03	0.05	588000.00	0.49	15.83	14.29	13.94	0.01	0.64	0.31	0.00	0.00	0.00	0.15	0.17	132.20
X15-393	259200.00	35.97	2433.00	0.01	0.02	0.03	0.06	589000.00	0.76	15.11	13.99	13.54	0.01	0.70	0.40	0.00	0.03	0.00	0.16	0.06	59.40
X15-393	264400.00	19.21	2494.00	0.01	0.01	0.04	0.06	604000.00	0.09	6.01	5.30	4.93	0.01	0.70	0.75	0.00	0.01	0.00	0.16	0.00	1.88
X15-393	253200.00	18.93	2150.00	0.01	0.00	0.03	0.06	579000.00	0.27	16.80	15.70	15.30	0.01	0.67	0.19	0.00	0.01	0.00	0.16	0.07	45.60
X15-393	251000.00	48.70	2851.00	0.02	0.10	0.02	0.05	544000.00	1.86	140.10	138.20	131.60	0.01	0.63	0.18	0.00	0.06	0.00	0.12	0.30	104.80
X15-393	261700.00	19.31	210.30	0.01	0.06	0.03	0.06	660000.00	9.57	26.51	25.84	24.60	0.02	1.07	0.85	0.00	0.30	0.00	0.32	0.54	164.90
X15-393	269900.00	6.05	386.00	0.03	0.04	0.07	0.09	660000.00	0.70	184.00	179.00	172.00	0.01	0.90	0.84	0.00	0.01	0.00	0.46	0.03	9.30
X15-393	256400.00	12.81	897.00	0.02	0.12	0.05	0.08	614000.00	3.79	47.47	47.20	44.27	0.02	0.76	0.22	0.00	0.08	0.00	0.27	0.02	116.00
X15-393	260000.00	22.61	472.10	0.01	0.06	0.04	0.09	643000.00	10.10	51.99	50.10	47.90	0.02	1.16	0.53	0.00	0.34	0.00	0.34	0.53	170.20
X15-393	263800.00	21.59	313.40	0.01	0.07	0.04	0.06	652000.00	11.24	47.30	44.90	44.40	0.02	1.15	0.74	0.00	0.40	0.00	0.34	0.52	172.90
X15-393	262300.00	13.50	286.90	0.01	0.11	0.05	0.07	652000.00	15.77	128.00	125.00	119.20	0.03	1.11	0.63	0.00	0.53	0.00	0.35	0.27	155.30
X15-393	267200.00	6.07	410.00	0.01	0.06	0.03	0.08	652000.00	12.25	188.00	186.30	177.60	0.03	0.98	0.42	0.00	0.44	0.00	0.33	0.04	92.50
X15-393	276500.00	6.34	447.40	0.01	0.11	0.06	0.10	651000.00	11.68	359.60	351.90	334.50	0.03	0.90	0.37	0.00	0.47	0.00	0.37	0.05	67.19
X15-393	297800.00	7.83	486.10	0.01	0.13	0.13	0.15	653000.00	12.52	417.60	406.20	389.90	0.03	0.89	0.42	0.00	0.52	0.00	0.48	0.07	175.00
X15-393	254700.00	7.80	467.30	0.02	0.17	0.08	0.11	624000.00	13.25	120.80	118.20	113.80	0.05	0.91	0.95	0.00	0.55	0.00	0.31	0.04	96.70
X15-393	267900.00	5.14	1024.30	0.01	-0.01	0.03	0.05	646000.00	0.00	35.93	33.20	32.91	0.01	0.75	0.41	0.00	0.00	0.00	0.27	0.00	0.73
X15-393	268100.00	0.47	90.60	0.03	-0.06	0.05	0.11	664000.00	0.03	14.10	12.60	12.60	0.01	1.05	3.21	0.00	0.00	0.00	0.30	0.00	0.16

W17-HT11	273300.00	439.50	5243.00	0.98	0.46	0.68	0.97	628000.00	13.96	24.05	22.02	22.14	0.04	0.84	0.84	0.00	0.76	0.00	-0.16	0.29	157.70
W17-HT11	261100.00	392.40	4946.00	0.93	1.60	0.46	0.56	599000.00	10.07	24.26	22.71	22.36	0.04	0.81	0.66	0.00	0.44	0.00	-0.44	0.29	87.60
W17-HT11	259100.00	24.50	406.60	0.21	-0.01	0.24	0.18	666000.00	0.07	16.93	17.15	15.36	0.01	0.68	3.07	0.00	0.00	0.00	0.08	0.38	54.37
W17-HT11	259000.00	29.70	479.00	0.20	0.02	0.60	0.50	670000.00	0.23	10.74	10.61	9.68	0.01	0.74	3.01	0.00	0.00	0.00	0.09	0.34	59.47
W17-HT11	254100.00	382.00	3565.00	0.61	1.03	0.56	0.47	597000.00	7.51	24.87	25.20	22.42	0.03	0.87	0.73	0.00	0.69	0.00	0.08	0.33	141.80
W17-HT11	257700.00	355.00	3380.00	0.53	0.46	0.81	0.59	595000.00	8.86	23.63	24.84	21.22	0.03	0.84	0.87	0.00	0.92	0.00	0.11	0.45	217.90
W17-HT11	263800.00	432.20	4382.00	0.71	1.05	0.97	0.69	623000.00	8.69	26.86	28.06	24.54	0.04	0.92	0.72	0.00	0.54	0.00	0.11	0.40	206.40
W17-HT11	252500.00	22.70	282.90	0.21	0.07	0.56	0.42	662000.00	0.39	11.47	11.86	10.53	0.01	0.71	3.31	0.00	0.03	0.00	0.31	0.29	77.50
W17-HT11	256100.00	18.35	291.80	0.14	0.06	0.18	0.24	667000.00	0.22	1.45	1.31	0.92	0.01	0.68	3.19	0.00	0.02	0.00	1.29	0.63	77.58
W17-HT11	252600.00	63.70	595.50	0.17	0.05	0.40	0.26	658000.00	0.89	4.75	4.57	3.90	0.01	0.68	2.04	0.00	0.05	0.00	-0.64	1.29	169.80
W17-HT11	254400.00	76.40	874.00	0.25	0.03	0.35	0.20	659000.00	0.26	5.35	5.11	4.51	0.01	0.68	1.12	0.00	0.01	0.00	-0.26	0.83	136.20
W17-HT11	254000.00	63.50	1373.00	1.24	1.08	3.43	1.35	660000.00	0.68	12.15	12.98	10.90	0.06	0.73	2.04	0.00	0.06	0.00	-0.13	0.76	181.30
W17-HT11	264100.00	47.60	5812.00	3.60	0.68	0.31	0.22	635000.00	0.08	24.66	26.27	22.05	0.01	0.67	0.32	0.00	0.00	0.00	-0.05	0.18	118.40
W17-HT11	241400.00	88.00	2504.00	1.65	0.07	1.76	0.58	615000.00	0.15	18.31	18.93	16.57	0.01	0.67	1.38	0.00	0.01	0.00	-0.06	0.43	128.20
W17-HT11	248900.00	64.60	4789.00	2.79	-0.04	0.27	0.29	611000.00	0.00	33.22	35.21	30.60	0.01	0.69	0.42	0.00	0.00	0.00	-0.04	0.06	451.10
W17-HT11	247400.00	136.70	4345.00	1.65	0.04	0.40	0.26	611000.00	0.01	18.35	19.03	16.58	0.01	0.66	1.60	0.00	0.00	0.00	-0.04	0.98	150.50
W17-HT11	251800.00	75.80	5325.00	3.13	0.09	0.69	0.22	611000.00	0.17	39.67	41.90	36.77	0.01	0.63	0.33	0.00	0.03	0.00	-0.03	0.08	319.00
W17-HT11	262400.00	436.20	5850.00	1.19	1.01	3.92	1.73	630000.00	12.16	29.84	31.81	27.99	0.03	1.01	0.71	0.00	0.64	0.00	-0.04	0.23	176.30
W17-HT11	244400.00	22.52	595.10	0.33	0.06	0.79	0.69	635000.00	0.16	31.17	33.76	29.42	0.01	0.66	2.51	0.00	0.00	0.00	-0.06	0.26	28.07
W17-HT11	255000.00	92.30	833.80	0.21	0.07	0.22	0.21	659000.00	0.31	3.60	3.38	3.01	0.01	0.70	3.13	0.00	0.02	0.00	-0.07	1.00	367.70
W17-HT11	254900.00	121.60	1818.00	0.27	0.06	0.17	0.16	647000.00	0.29	4.44	4.15	3.74	0.01	0.68	2.78	0.00	0.00	0.00	-0.08	1.40	397.00
W17-HT11	257000.00	120.70	1547.00	0.26	0.03	0.13	0.17	651000.00	0.55	5.80	5.81	5.07	0.01	0.68	2.34	0.00	0.00	0.00	-0.12	1.41	469.00
W17-HT11	260100.00	110.40	899.80	0.21	-0.01	0.10	0.12	658000.00	0.10	4.26	4.04	3.73	0.01	0.69	2.29	0.00	0.01	0.00	-0.21	1.26	232.80
W17-HT12	344500.00	99.90	7180.00	2.85	42.00	0.13	0.12	642000.00	0.20	38.20	41.90	36.10	0.01	0.69	0.30	0.00	0.00	0.00	-0.07	0.13	64.90
W17-HT12	304200.00	55.90	4447.00	3.00	53.10	0.70	0.70	598000.00	1.30	34.01	37.20	32.37	0.02	0.68	0.47	0.00	0.06	0.00	-0.08	0.11	58.90
W17-HT12	308700.00	62.50	5224.00	3.44	0.47	0.07	0.10	604000.00	0.46	29.07	31.90	27.16	0.01	0.71	0.14	0.00	0.03	0.00	-0.06	0.05	116.40
W17-HT12	317500.00	125.00	4858.00	2.08	0.02	0.04	0.08	635000.00	0.03	29.73	31.81	28.33	0.01	0.74	0.62	0.00	0.00	0.00	-0.06	0.19	70.90
W17-HT12	302600.00	103.60	2758.00	1.54	-0.03	0.04	0.09	640000.00	0.02	41.24	44.63	39.32	0.01	0.78	1.74	0.00	0.00	0.00	-0.06	0.33	1153.00
W17-HT12	310000.00	302.00	3215.00	0.52	0.88	1.11	1.06	618000.00	8.86	31.77	33.53	30.27	0.01	1.03	1.75	0.00	1.22	0.00	-0.08	0.37	162.20
W17-HT12	286600.00	73.40	3779.00	0.29	0.06	0.10	0.13	599000.00	0.65	9.10	8.98	8.16	0.01	0.78	1.40	0.00	0.13	0.00	-0.06	0.34	371.60
W17-HT12	305900.00	162.00	2792.00	0.60	0.08	0.06	0.11	658000.00	0.78	18.71	19.57	17.86	0.01	0.94	1.69	0.00	0.02	0.00	-0.08	0.86	536.00
W17-HT12	289800.00	29.86	465.90	0.17	0.02	0.06	0.10	658000.00	0.43	2.85	2.61	2.41	0.01	0.91	2.98	0.00	0.00	0.00	-0.12	0.88	170.90
W17-HT12	286600.00	88.80	847.00	0.16	0.06	0.26	0.26	651000.00	1.22	13.93	13.76	13.27	0.01	0.90	2.82	0.00	0.22	0.00	-0.16	1.51	412.80
W17-HT12	286700.00	98.10	872.00	0.11	-0.03	0.53	0.54	651000.00	2.60	14.46	13.99	13.62	0.01	0.90	2.77	0.00	0.41	0.00	-0.20	0.43	1439.00
W17-HT12	280800.00	49.70	450.50	0.19	0.02	0.15	0.17	659000.00	0.57	3.04	2.72	2.61	0.01	0.81	2.71	0.00	0.07	0.00	-0.45	0.63	184.50
W17-HT12	261600.00	9.34	208.30	0.28	0.00	-0.25	0.28	650000.00	0.70	1.06	0.57	0.59	0.02	0.52	3.79	0.00	0.06	0.00	0.08	0.37	46.30
W17-HT12	293100.00	822.40	9560.00	2.19	1.12	2.63	3.95	659000.00	20.06	25.60	24.41	22.93	0.07	0.71	1.00	0.00	0.73	0.00	0.06	0.51	336.40
W17-HT12	266100.00	634.00	6886.00	2.13	0.55	2.24	3.16	619000.00	23.11	22.86	22.38	20.36	0.04	0.64	1.10	0.00	1.07	0.00	0.07	0.22	214.20
W17-HT12	265000.00	27.30	745.00	0.50	-0.04	0.43	0.76	660000.00	3.08	5.73	5.07	4.80	0.02	0.51	3.32	0.00	0.11	0.00	0.12	0.38	51.61
W17-HT12	262200.00	9.50	349.00	0.48	0.04	9.40	14.30	662000.00	1.35	2.44	1.89	1.95	0.02	0.63	4.26	0.00	0.00	0.00	0.14	0.08	12.86
W17-HT12	259300.00	0.78	67.50	0.23	-0.19	51.10	75.30	662000.00	21.80	0.83	0.39	0.40	0.02	0.58	4.10	0.00	0.01	0.00	0.18	0.06	6.40
W17-HT12	257700.00	267.80	8690.00	2.11	-0.10	0.20	0.28	598000.00	0.08	15.16	13.97	13.39	0.03	0.43	0.29	0.00	0.00	0.00	0.19	0.01	60.90
W17-HT12	272100.00	205.00	7268.00	0.80	1.05	0.72	1.13	634000.00	8.34	14.17	12.82	12.69	0.02	0.53	1.13	0.00	0.39	0.00	1.06	0.31	186.20
W17-HT12	271100.00	135.50	7449.00	0.61	-0.11	0.12	0.41	638000.00	2.07	7.95	7.20	6.74	0.02	0.51	1.37	0.00	0.12	0.00	-0.53	0.47	191.30
W17-HT12	275200.00	590.00	6610.00	1.20	3.60	2.79	4.02	629000.00	15.84	27.61	25.47	25.17	0.03	0.69	1.56	0.00	0.65	0.00	-0.26	0.46	112.40
W17-HT12	265600.00	16.66	438.40	0.40	-0.05	0.11	0.32	656000.00	0.05	2.49	2.02	1.95	0.02	0.53	4.26	0.00	0.00	0.00	-0.22	0.51	58.20
W17-HT12	267700.00	34.71	715.40	0.33	0.05	0.10	0.20	662000.00	0.89	1.67	1.12	1.16	0.02	0.58	3.42	0.00	0.07	0.00	-0.17	0.50	89.30
W17-HT12	269400.00	33.93	758.10	0.32	-0.01	0.07	0.18	661000.00	0.81	2.22	1.49	1.74	0.01	0.59	2.95	0.00	0.01	0.00	-0.14	0.68	99.60
W17-HT12	271200.00	36.10	1228.00	0.60	0.02	0.19	0.25	668000.00	0.66	37.10	35.20	34.90	0.02	0.59	2.96	0.00	0.05	0.00	-0.13	0.67	127.00
W17-HT12	275000.00	87.60	1649.00	0.76	-0.03	0.08	0.17	670000.00	0.17	48.00	45.10	44.60	0.01	0.60	0.99	0.00	0.02	0.00	-0.12	0.26	325.00
W17-HT12	272700.00	69.00	1828.00	1.66	-0.09	0.09	0.19	665000.00	0.06	10.96	9.81	9.78	0.01	0.61	2.92	0.00	0.00	0.00	-0.13	1.11	182.60
W17-HT12	264900.00	34.34	978.00	1.16	0.05	2.42	2.71	650000.00	0.94	4.58	4.01	3.85	0.01	0.65	3.52	0.00	0.06	0.00	-0.16	0.65	101.80

X15-28	233900.00	38.01	3231.00	0.37	0.02	-0.06	0.07	614000.00	10.22	88.40	85.60	82.90	0.01	0.61	0.81	0.00	0.19	0.00	0.17	0.11	53.90
X15-28	235000.00	34.50	2982.00	0.28	0.10	0.05	0.12	632000.00	7.23	79.26	76.50	74.90	0.01	0.71	0.92	0.00	0.11	0.00	0.18	0.10	62.10
X15-28	257200.00	242.00	10180.00	1.98	0.07	0.02	0.14	629000.00	1.81	108.40	103.70	101.50	0.01	0.77	0.26	0.00	0.04	0.00	0.08	0.29	375.00
X15-28	226300.00	72.40	4610.00	2.40	0.06	0.04	0.20	608000.00	0.02	22.86	21.57	21.22	0.01	0.77	0.08	0.00	0.00	0.00	0.12	0.20	200.00
X15-28	216500.00	105.40	3635.00	1.31	0.03	0.03	0.13	591000.00	0.89	44.80	42.60	42.10	0.01	0.83	0.64	0.00	0.03	0.00	0.16	1.49	433.00
X15-28	230600.00	82.10	3517.00	0.40	0.00	0.05	0.17	637000.00	4.50	85.90	81.60	81.30	0.01	0.87	0.85	0.00	0.06	0.00	0.19	0.23	285.00
X15-28	225300.00	29.41	1859.00	0.08	0.05	0.09	0.17	642000.00	6.34	42.42	39.95	40.56	0.01	0.97	1.03	0.00	0.13	0.00	0.25	0.28	207.00
X15-28	217900.00	105.70	3377.00	1.20	-0.01	0.06	0.16	607000.00	0.62	47.70	44.47	45.26	0.01	0.91	0.67	0.00	0.01	0.00	0.22	1.62	561.00
X15-28	220300.00	37.26	2354.00	0.16	0.01	0.05	0.22	629000.00	4.96	75.30	70.60	71.40	0.01	1.07	1.04	0.00	0.11	0.00	0.32	0.13	104.80
X15-28	239800.00	22.66	1380.00	0.07	0.17	1.40	0.66	625000.00	8.67	40.58	39.13	38.84	0.01	1.06	1.18	0.00	0.29	0.00	0.60	0.24	184.10
X15-28	218900.00	101.40	3051.00	1.07	0.00	0.03	0.14	632000.00	1.47	51.27	48.90	49.54	0.01	1.08	0.84	0.00	0.04	0.00	0.36	1.64	657.00
X15-28	225900.00	152.00	6350.00	0.82	-0.01	0.03	0.13	598000.00	0.01	58.20	54.00	56.00	0.01	1.10	0.52	0.00	0.00	0.00	0.36	0.31	60.30
X15-28	220200.00	79.10	3877.00	1.46	0.00	0.06	0.18	613000.00	0.07	44.24	40.76	42.25	0.01	1.08	0.88	0.00	0.00	0.00	0.99	0.15	61.10
X15-28	217200.00	65.00	1130.00	0.21	-0.03	0.08	0.18	636000.00	3.32	28.29	26.06	27.21	0.01	1.14	1.09	0.00	0.11	0.00	14.60	0.64	514.00
X15-28	221600.00	80.60	1290.00	0.24	0.00	0.06	0.17	646000.00	3.74	28.68	27.02	27.46	0.01	1.21	1.04	0.00	0.09	0.00	-3.14	0.77	665.00
X15-28	217800.00	71.80	990.00	0.25	0.04	0.08	0.17	640000.00	2.07	27.13	25.34	26.19	0.01	1.08	1.24	0.00	0.01	0.00	-1.53	0.71	470.00
X15-28	209400.00	69.50	2322.00	0.48	0.01	0.04	0.16	592000.00	0.06	22.60	20.80	21.40	0.02	1.00	0.83	0.00	0.00	0.00	-0.77	0.04	19.61
X15-28	227700.00	78.20	4450.00	1.47	0.42	0.06	0.14	609000.00	0.19	36.84	34.80	35.60	0.02	1.03	0.39	0.00	0.01	0.00	-0.53	0.17	75.80
X15-28	232400.00	43.06	2320.00	0.84	0.05	0.07	0.19	645000.00	0.65	33.20	30.90	32.20	0.01	1.03	0.60	0.00	0.02	0.00	-1.09	0.07	22.60
X15-28	246200.00	116.90	5660.00	1.56	0.05	0.06	0.12	639000.00	0.01	31.30	28.87	29.62	0.01	1.03	0.54	0.00	0.00	0.00	-0.96	0.44	191.00
X15-28	230200.00	148.00	4630.00	1.34	0.04	0.05	0.06	600000.00	0.25	46.50	44.00	44.70	0.01	0.82	0.48	0.00	0.00	0.00	-4.94	0.28	545.00
X15-28	215500.00	127.60	4570.00	1.26	0.03	0.05	0.15	553000.00	0.39	44.46	42.90	42.76	0.01	0.72	0.39	0.00	0.00	0.00	0.96	0.33	229.10
X15-28	231600.00	154.70	6514.00	1.30	0.04	0.06	0.13	565000.00	0.61	99.23	95.10	96.50	0.01	0.70	0.38	0.00	0.00	0.00	0.29	0.26	822.00
X15-28	296900.00	119.80	5735.00	1.69	0.04	-0.05	0.03	618000.00	0.02	16.18	16.53	14.49	0.01	0.46	1.30	0.00	0.00	0.00	0.09	1.17	173.70
X15-28	304400.00	153.30	5736.00	1.64	0.13	0.05	0.05	621000.00	0.59	20.35	21.68	18.37	0.01	0.48	0.56	0.00	0.02	0.00	0.10	0.18	94.10
X15-28	318900.00	67.50	5920.00	0.77	-0.10	0.11	0.11	639000.00	0.01	85.90	95.30	79.20	0.01	0.45	0.32	0.00	0.00	0.00	0.12	0.10	9.65
X15-28	311300.00	150.20	5223.00	1.42	0.14	0.21	0.09	610000.00	1.38	110.10	124.00	101.00	0.02	0.46	0.26	0.00	0.03	0.00	0.16	0.28	118.50
X15-28	322900.00	115.40	5840.00	0.73	0.15	0.06	0.13	629000.00	1.66	103.50	115.90	96.00	0.01	0.53	0.65	0.00	0.03	0.00	0.20	0.19	225.00
X15-28	318300.00	42.80	2751.00	0.29	0.08	0.25	0.26	619000.00	2.58	58.65	66.20	54.40	0.02	0.50	0.82	0.00	0.07	0.00	0.52	0.29	135.00
X15-28	311500.00	210.90	4975.00	1.35	0.09	0.06	0.12	615000.00	0.17	38.50	42.90	35.80	0.01	0.47	0.65	0.00	0.01	0.00	0.66	0.57	566.00
X15-28	316700.00	51.50	1734.00	0.13	0.12	0.07	0.14	629000.00	5.37	37.40	42.60	34.55	0.01	0.52	1.08	0.00	0.18	0.00	-2.04	0.57	354.90
X15-28	360600.00	33.34	1811.00	0.06	0.27	0.07	0.13	645000.00	15.42	61.15	68.50	57.84	0.01	0.58	1.36	0.00	0.69	0.00	-1.28	0.28	152.70
X15-28	369500.00	65.70	1751.00	0.10	0.99	2.80	2.17	645000.00	10.17	57.71	64.90	53.81	0.04	0.61	1.28	0.00	0.43	0.00	-0.79	0.25	194.90
X15-28	368900.00	147.20	1836.00	0.08	0.73	0.69	0.66	645000.00	12.27	69.50	76.10	66.00	0.03	0.61	1.34	0.00	0.47	0.00	-0.65	0.28	174.50
X15-28	344800.00	31.32	1714.00	0.26	0.19	0.20	0.21	645000.00	6.48	46.22	51.50	44.47	0.01	0.67	1.05	0.00	0.25	0.00	-0.19	0.35	175.00

K57BF	243500.00	73.40	2344.00	0.25	0.02	-0.01	0.13	628000.00	0.13	106.80	107.90	101.50	0.01	0.42	1.13	0.00	0.01	0.00	0.26	0.20	46.40
K57BF	241700.00	87.30	3049.00	0.36	0.02	0.07	0.19	618000.00	0.05	180.10	182.30	170.50	0.01	0.41	0.94	0.00	0.00	0.00	0.21	0.40	118.70
K57BF	251600.00	17.75	1042.00	0.42	0.06	0.10	0.26	670000.00	0.00	45.29	45.60	42.77	0.01	0.46	1.02	0.00	0.00	0.00	0.28	0.05	91.00
K57BF	249800.00	37.52	1966.00	0.43	0.04	0.07	0.19	656000.00	0.00	57.40	58.30	54.40	0.01	0.45	0.87	0.00	0.00	0.00	0.24	0.06	119.00
K57BF	247700.00	6.41	437.20	0.23	0.01	0.28	0.34	664000.00	0.00	9.83	9.27	8.95	0.01	0.48	2.43	0.00	0.00	0.00	0.27	0.01	24.31
K57BF	250400.00	15.29	938.00	0.28	0.00	0.07	0.20	667000.00	0.00	16.19	15.85	15.29	0.01	0.46	0.77	0.00	0.00	0.00	0.25	0.01	172.70
K57BF	247000.00	70.97	1163.00	0.16	0.03	0.12	0.20	654000.00	0.00	27.89	27.67	26.16	0.01	0.49	3.09	0.00	0.00	0.00	0.24	0.22	102.40
K57BF	248400.00	56.30	3193.00	0.22	0.07	0.12	0.22	634000.00	0.01	49.40	50.30	47.20	0.01	0.48	2.13	0.00	0.00	0.00	0.18	0.14	190.00
K57BF	249200.00	97.30	3334.00	0.25	0.08	0.50	0.58	632000.00	1.27	196.20	201.50	189.00	0.01	0.57	1.16	0.00	0.02	0.00	0.16	0.78	381.10
K57BF	247600.00	108.70	3427.00	0.20	0.06	0.06	0.34	631000.00	0.00	210.10	215.60	200.50	0.01	0.51	1.03	0.00	0.00	0.00	0.15	0.38	129.80
K57BF	246400.00	51.52	4228.00	0.28	0.05	0.07	0.16	619000.00	0.01	72.80	73.21	69.99	0.01	0.52	1.12	0.00	0.00	0.00	0.14	0.14	51.12
K57BF	247100.00	41.52	1245.50	0.36	0.06	0.15	0.24	651000.00	0.00	83.25	85.20	80.16	0.01	0.57	1.19	0.00	0.00	0.00	0.21	0.11	46.31
K57BF	252800.00	71.69	2108.00	0.24	0.06	0.08	0.19	653000.00	0.00	149.90	154.10	144.90	0.01	0.60	0.69	0.00	0.00	0.00	0.19	0.23	70.40
K57BF	250600.00	73.80	746.70	0.12	0.02	0.11	0.24	660000.00	0.03	23.52	23.82	22.20	0.01	0.62	6.48	0.00	0.00	0.00	0.21	0.29	281.20
K57BF	249800.00	55.77	2202.00	0.19	0.03	0.12	0.24	643000.00	0.16	33.51	34.01	31.80	0.01	0.63	2.25	0.00	0.00	0.00	0.18	0.26	175.00
K57BF	254700.00	56.24	1204.80	0.14	0.01	0.08	0.18	664000.00	0.00	21.40	21.20	20.28	0.01	0.66	2.85	0.00	0.00	0.00	0.22	0.12	64.91
K57BF	249700.00	106.10	3661.00	0.18	0.05	0.07	0.16	619000.00	0.14	252.40	257.20	242.50	0.01	0.67	0.88	0.00	0.00	0.00	0.14	0.59	121.70
K57BF	253100.00	77.63	875.80	0.16	0.02	0.09	0.20	658000.00	0.00	32.49	32.46	30.73	0.01	0.69	2.95	0.00	0.00	0.00	0.21	0.35	232.90
K57BF	260600.00	57.80	1345.40	0.21	0.04	0.09	0.21	662000.00	0.03	56.26	57.20	53.38	0.01	0.70	1.14	0.00	0.00	0.00	0.20	0.19	518.00
K57BF	278400.00	40.26	1724.00	0.24	-0.01	-0.06	0.01	654000.00	0.00	56.00	54.78	51.76	0.01	0.48	0.23	0.00	0.00	0.00	0.21	0.03	440.00
K57BF	282200.00	126.60	731.80	0.07	0.01	0.12	0.24	663000.00	0.00	22.49	21.95	20.47	0.01	0.47	5.90	0.00	0.00	0.00	0.25	0.28	196.80
K57BF	281900.00	81.61	1611.00	0.08	0.05	0.11	0.15	649000.00	0.07	62.20	60.60	57.00	0.01	0.44	3.85	0.00	0.00	0.00	0.22	0.51	470.00
K57BF	286100.00	75.66	1499.00	0.11	0.03	0.10	0.19	659000.00	0.00	54.17	52.81	49.82	0.01	0.44	3.00	0.00	0.00	0.00	0.21	0.28	486.00
K57BF	288800.00	63.65	1411.00	0.10	0.03	0.08	0.20	659000.00	0.03	69.77	67.40	64.32	0.01	0.44	1.95	0.00	0.00	0.00	0.22	0.33	129.30
K57BF	291300.00	73.80	748.90	0.10	0.06	0.35	0.38	665000.00	0.08	58.89	56.97	54.13	0.01	0.44	4.60	0.00	0.00	0.00	0.23	0.30	135.90
K57BF	288000.00	64.90	1951.00	0.13	0.03	0.08	0.17	648000.00	0.00	120.90	118.40	112.00	0.01	0.40	0.78	0.00	0.00	0.00	0.18	0.16	471.00
K57BF	294200.00	71.60	1099.00	0.11	0.01	0.63	0.66	667000.00	0.01	48.26	47.10	44.16	0.01	0.46	5.52	0.00	0.00	0.00	0.22	0.25	193.10
K57BF	293700.00	55.50	2151.00	0.14	0.03	0.07	0.19	654000.00	0.01	60.07	59.15	54.98	0.01	0.41	2.30	0.00	0.00	0.00	0.19	0.23	91.30
K57BF	290400.00	6.03	507.40	0.24	0.02	0.10	0.20	659000.00	0.01	37.80	36.54	34.47	0.01	0.40	3.70	0.00	0.00	0.00	0.24	0.02	26.14
K57BF	300000.00	69.90	2141.00	0.30	0.05	0.10	0.16	663000.00	0.00	109.30	107.10	101.30	0.01	0.42	2.46	0.00	0.00	0.00	0.22	0.12	57.60
K57BF	286400.00	135.90	4457.00	0.33	0.11	0.13	0.16	614000.00	0.12	240.50	237.70	222.40	0.01	0.38	2.17	0.00	0.00	0.00	0.16	0.44	119.60
K57BF	293400.00	3.86	249.60	0.19	0.00	6.87	5.89	669000.00	0.01	9.60	9.24	8.37	0.01	0.73	6.51	0.00	0.00	0.00	0.23	0.03	13.47
K57BF	292600.00	9.68	473.30	0.21	0.02	0.10	0.22	670000.00	0.00	12.01	11.43	10.77	0.01	0.42	4.87	0.00	0.00	0.00	0.21	0.05	30.78
K57BF	289100.00	136.10	3090.00	0.26	0.02	0.09	0.16	643000.00	0.00	405.20	401.30	375.40	0.01	0.41	0.76	0.00	0.00	0.00	0.17	0.89	239.30
K57BF	290700.00	23.36	594.20	0.14	0.01	3.06	2.91	667000.00	0.01	9.19	8.73	8.13	0.01	0.48	6.98	0.00	0.00	0.00	0.22	0.05	39.18
K57BF	288200.00	47.64	1515.00	0.16	0.00	0.16	0.26	662000.00	0.01	31.80	31.29	28.99	0.01	0.43	3.55	0.00	0.00	0.00	0.22	0.12	81.90
K57BF	288200.00	2.18	279.80	0.24	0.02	17.96	16.61	674000.00	0.01	2.03	1.66	1.47	0.01	0.49	5.69	0.00	0.00	0.00	0.21	0.00	7.80
K57BF	285000.00	21.82	1272.00	0.35	0.07	0.68	0.68	663000.00	0.00	18.58	17.98	16.94	0.01	0.46	1.13	0.00	0.00	0.00	0.20	0.01	244.60
K57BF	277700.00	113.30	3637.00	0.36	0.08	0.45	0.57	631000.00	0.00	258.70	255.30	241.80	0.01	0.53	0.32	0.00	0.00	0.00	0.15	0.55	202.20
K57BF	249800.00	58.90	2960.00	0.25	0.05	0.15	0.23	609000.00	0.00	86.20	86.00	79.30	0.01	0.85	0.34	0.00	0.00	0.00	0.15	0.11	46.60
K57BF	257300.00	6.50	294.80	0.23	0.01	2.55	3.17	670000.00	0.03	10.22	10.02	8.95	0.01	1.27	3.61	0.00	0.00	0.00	0.24	0.02	14.43
K57BF	255900.00	40.90	673.10	0.20	0.01	0.22	0.29	659000.00	0.00	28.77	28.27	26.45	0.01	1.01	3.42	0.00	0.00	0.00	0.23	0.10	82.50

O28	254000.00	54.98	1549.00	0.08	0.03	0.87	1.04	640000.00	0.51	32.91	33.11	30.27	0.01	0.91	5.14	0.00	0.01	0.00	0.21	0.15	260.00
O28	256500.00	52.10	1649.00	0.06	0.02	0.35	0.48	642000.00	0.24	34.70	34.28	31.83	0.01	0.96	5.66	0.00	0.00	0.00	0.19	0.16	287.00
O28	264000.00	16.59	1897.00	0.21	0.06	0.28	0.35	657000.00	0.02	18.29	17.83	16.46	0.01	0.90	1.64	0.00	0.00	0.00	0.20	0.03	19.34
O28	259300.00	58.79	468.40	0.13	0.03	0.66	0.86	666000.00	0.20	13.43	13.00	12.08	0.01	1.13	2.82	0.00	0.00	0.00	0.21	0.01	44.91
O28	262300.00	14.80	1319.00	0.21	0.03	0.29	0.38	657000.00	0.07	36.70	36.10	33.86	0.01	0.88	1.18	0.00	0.00	0.00	0.19	0.04	25.95
O28	254900.00	4.85	201.40	0.18	0.00	0.08	0.16	656000.00	0.01	15.54	15.49	14.18	0.01	0.89	3.57	0.00	0.00	0.00	0.21	0.03	30.73
O28	258400.00	36.37	892.00	0.14	0.02	0.11	0.17	653000.00	0.02	32.00	31.30	29.40	0.01	0.85	1.59	0.00	0.00	0.00	0.19	0.03	29.88
O28	253500.00	1.43	128.90	0.19	0.01	3.72	4.34	645000.00	0.05	2.33	1.94	1.82	0.01	1.00	3.45	0.00	0.00	0.00	0.20	0.00	8.09
O28	254000.00	20.12	956.70	0.17	0.01	0.04	0.09	630000.00	0.00	36.02	36.09	33.29	0.01	0.72	1.93	0.00	0.00	0.00	0.20	0.15	91.44
O28	266700.00	43.49	1518.00	0.16	0.01	0.04	0.10	649000.00	0.00	30.82	30.67	28.55	0.01	0.73	2.77	0.00	0.00	0.00	0.18	0.09	101.10
O28	260200.00	30.20	591.40	0.17	0.01	0.04	0.08	648000.00	0.00	48.86	48.34	45.82	0.01	0.73	1.53	0.00	0.00	0.00	0.17	0.01	734.70
O28	264000.00	44.22	1810.00	0.14	0.00	0.03	0.07	632000.00	0.03	34.51	33.78	31.92	0.01	0.67	2.78	0.00	0.00	0.00	0.16	0.14	183.00
O28	266900.00	50.79	2325.00	0.19	0.01	0.05	0.09	627000.00	0.16	65.17	64.38	60.80	0.01	0.65	1.41	0.00	0.01	0.00	0.16	0.07	312.10
O28	271000.00	47.41	2838.00	0.22	-0.02	0.04	0.07	628000.00	0.00	30.99	30.33	28.27	0.01	0.65	1.36	0.00	0.00	0.00	0.14	0.05	59.70
O28	263600.00	41.88	508.40	0.15	0.00	0.04	0.09	650000.00	0.08	18.84	18.28	17.51	0.01	0.62	4.11	0.00	0.00	0.00	0.20	0.12	83.90
O28	265200.00	44.72	663.20	0.15	0.03	0.04	0.07	654000.00	0.00	22.38	21.91	20.53	0.01	0.63	3.92	0.00	0.00	0.00	0.20	0.11	85.00
O28	267300.00	39.40	535.90	0.18	0.03	0.04	0.10	661000.00	0.02	20.68	20.65	19.16	0.01	0.63	3.64	0.00	0.00	0.00	0.20	0.07	61.30
O28	264100.00	85.47	1919.00	0.24	0.03	-0.05	0.02	658000.00	0.04	279.60	279.80	260.40	0.01	0.56	0.09	0.00	0.00	0.00	0.16	0.38	198.50
O28	267100.00	14.72	1484.00	0.21	0.02	0.03	0.10	656000.00	0.00	25.52	25.00	23.36	0.00	0.55	0.42	0.00	0.00	0.00	0.16	0.05	23.76
O28	263800.00	24.10	1062.10	0.15	0.03	0.02	0.05	656000.00	0.00	20.25	19.40	18.33	0.00	0.54	0.75	0.00	0.00	0.00	0.17	0.04	37.83
O28	267800.00	64.52	1911.00	0.22	0.01	0.02	0.03	627000.00	0.02	433.60	432.50	398.80	0.00	0.51	0.10	0.00	0.00	0.00	0.14	0.73	182.70
O28	257600.00	23.04	337.90	0.09	0.00	0.04	0.05	640000.00	0.00	28.16	27.70	25.61	0.00	0.50	1.20	0.00	0.00	0.00	0.20	0.16	118.30
O28	263400.00	21.66	921.00	0.13	0.00	0.02	0.04	628000.00	0.00	46.97	46.80	42.60	0.00	0.49	0.17	0.00	0.00	0.00	0.15	0.11	85.50
O28	265700.00	28.07	290.10	0.06	0.01	0.02	0.05	655000.00	0.00	27.04	26.43	24.32	0.00	0.50	1.03	0.00	0.00	0.00	0.19	0.25	159.80
O28	262800.00	19.76	399.00	0.07	0.00	0.02	0.06	634000.00	0.00	43.20	42.16	39.59	0.00	0.47	0.51	0.00	0.00	0.00	0.18	0.23	187.00
O28	277300.00	25.08	781.00	0.13	0.00	0.01	0.03	642000.00	0.01	192.30	189.90	176.50	0.00	0.49	0.06	0.00	0.00	0.00	0.14	0.26	106.60
O28	275500.00	6.69	122.50	0.05	-0.01	0.02	0.04	659000.00	0.00	17.03	16.41	15.27	0.00	0.51	1.08	0.00	0.00	0.00	0.18	0.11	67.27
O28	253400.00	19.58	694.50	0.07	0.02	0.01	0.03	570000.00	0.00	121.50	118.30	110.20	0.00	0.44	0.07	0.00	0.00	0.00	0.12	0.16	75.40
O28	330500.00	51.32	2088.00	0.04	0.02	0.01	0.03	659000.00	0.00	669.00	657.00	614.00	0.00	0.53	0.04	0.00	0.00	0.00	0.11	1.88	232.70
O28	288400.00	4.41	163.00	0.04	0.01	0.02	0.04	668000.00	0.00	48.90	47.70	44.48	0.00	0.56	0.15	0.00	0.00	0.00	0.16	0.03	155.30
W17-46	282700.00	36.15	193.60	0.02	0.02	0.03	0.07	650000.00	1.33	16.51	15.82	14.58	0.00	0.60	1.05	0.00	0.11	0.00	0.13	3.22	1698.00
W17-46	284900.00	42.87	621.50	0.05	0.01	0.01	0.03	630000.00	0.27	21.97	21.15	19.69	0.00	0.54	0.08	0.00	0.01	0.00	0.09	1.17	748.00
W17-46	297000.00	38.60	1093.00	0.09	0.01	0.02	0.03	626000.00	0.05	34.43	33.90	31.07	0.00	0.48	0.05	0.00	0.00	0.00	0.07	0.06	226.80
W17-46	291600.00	0.31	65.55	0.02	0.01	0.02	0.03	667000.00	0.00	11.91	11.43	10.66	0.00	0.55	0.63	0.00	0.00	0.00	0.13	0.00	1.34
W17-46	287300.00	23.99	130.21	0.01	0.02	0.02	0.05	655000.00	2.13	16.53	15.88	14.68	0.00	0.57	2.17	0.00	0.20	0.00	0.15	1.90	1188.00
W17-46	286700.00	38.04	253.40	0.02	0.01	0.02	0.04	654000.00	0.50	13.60	12.73	12.14	0.00	0.54	0.71	0.00	0.03	0.00	0.14	3.22	1176.00
W17-46	285700.00	37.16	1257.20	0.11	0.01	0.02	0.04	614000.00	0.02	22.92	22.38	20.30	0.00	0.49	0.06	0.00	0.00	0.00	0.10	0.07	107.80
W17-46	297300.00	0.62	134.50	0.01	0.02	0.03	0.05	668000.00	0.00	2.88	2.35	2.23	0.00	0.53	1.06	0.00	0.00	0.00	0.28	0.01	2.19

N38A	242100.00	150.30	5185.00	0.13	0.13	0.14	0.28	630000.00	0.75	37.20	36.57	34.94	0.02	0.44	0.87	0.00	0.00	0.00	0.13	0.12	219.90
N38A	234300.00	146.50	2523.00	0.15	0.04	0.35	0.46	627000.00	5.45	76.70	76.10	72.70	0.02	0.56	0.67	0.00	0.01	0.00	0.25	0.43	664.20
N38A	222400.00	127.60	2183.00	0.14	0.05	0.67	0.76	603000.00	4.54	35.00	34.80	32.92	0.02	0.51	1.47	0.00	0.02	0.00	0.23	0.59	782.00
N38A	231800.00	161.50	2913.00	0.14	0.04	0.22	0.33	626000.00	4.74	40.26	39.58	37.69	0.02	0.56	1.36	0.00	0.03	0.00	0.20	0.64	962.00
N38A	233400.00	164.00	2730.00	0.13	0.04	0.12	0.22	628000.00	5.18	42.39	41.99	39.70	0.02	0.56	1.59	0.00	0.03	0.00	0.19	0.72	1095.00
N38A	223200.00	120.00	3292.00	0.13	0.07	0.81	0.87	583000.00	4.65	89.80	89.00	84.60	0.03	0.56	1.11	0.00	0.02	0.00	0.15	0.32	522.00
N38A	228800.00	129.00	3978.00	0.11	0.05	3.05	2.98	599000.00	4.69	29.19	29.22	27.12	0.02	0.51	1.40	0.00	0.03	0.00	0.15	0.45	565.70
N38A	256900.00	168.80	4641.00	0.13	0.08	2.20	2.21	662000.00	4.77	39.12	38.97	36.69	0.03	0.51	0.84	0.00	0.02	0.00	0.16	0.27	571.20
N38A	240800.00	73.80	1005.00	0.09	0.02	8.69	8.78	657000.00	9.16	30.06	29.74	28.13	0.02	0.53	2.01	0.00	0.08	0.00	0.27	0.56	617.20
N38A	239500.00	60.90	797.00	0.07	0.00	0.09	0.18	657000.00	0.00	5.30	4.89	4.61	0.01	0.45	1.49	0.00	0.00	0.00	0.26	0.24	81.60
N38A	238800.00	2.26	119.30	0.09	0.01	0.08	0.14	662000.00	0.01	0.80	0.43	0.39	0.01	0.47	4.15	0.00	0.00	0.00	0.24	0.12	28.86
N38A	251300.00	46.35	4543.00	0.30	0.02	0.06	0.18	646000.00	0.00	47.88	47.05	44.90	0.01	0.45	0.25	0.00	0.00	0.00	0.14	0.04	15.91
N38A	244700.00	90.40	1704.00	0.13	0.01	0.09	0.20	660000.00	0.00	58.71	58.30	55.35	0.01	0.47	0.41	0.00	0.00	0.00	0.20	0.03	1028.00
N38A	239100.00	50.70	881.60	0.14	0.03	0.08	0.20	652000.00	0.00	35.59	35.24	33.24	0.01	0.46	1.31	0.00	0.00	0.00	0.23	0.10	42.32
N38A	240400.00	57.70	973.60	0.12	0.01	0.08	0.18	652000.00	0.00	34.73	34.36	32.57	0.01	0.46	2.07	0.00	0.00	0.00	0.22	0.13	115.00
N38A	233700.00	30.44	506.70	0.10	0.01	0.08	0.17	635000.00	0.10	10.43	10.27	9.32	0.01	0.47	3.13	0.00	0.00	0.00	0.22	0.10	67.90
N38A	252700.00	72.70	3030.00	0.16	0.01	0.08	0.20	657000.00	0.00	37.67	37.10	35.17	0.01	0.50	0.79	0.00	0.00	0.00	0.17	0.04	297.00
N38A	234000.00	16.01	356.90	0.07	0.02	1.06	1.14	631000.00	0.37	2.59	2.08	2.01	0.01	0.49	5.47	0.00	0.00	0.00	0.21	0.02	11.59
N38A	235100.00	104.50	755.00	0.07	0.03	5.79	5.90	625000.00	12.59	31.58	31.34	29.17	0.02	0.59	2.22	0.00	0.12	0.00	0.21	0.84	949.00
N38A	217300.00	121.60	945.00	0.06	0.03	3.74	3.94	572000.00	16.09	34.39	33.94	31.72	0.02	0.60	1.98	0.00	0.16	0.00	0.20	0.84	1020.00
N38A	245200.00	171.00	3634.00	0.16	0.08	0.11	0.16	617000.00	2.46	45.07	45.28	41.63	0.02	0.50	0.98	0.00	0.01	0.00	0.14	0.19	528.00
N38A	243900.00	176.90	3883.00	0.16	0.04	0.10	0.19	606000.00	4.28	41.25	40.86	38.08	0.02	0.59	0.88	0.00	0.02	0.00	0.14	0.33	653.30
N38A	254200.00	182.60	2817.00	0.11	0.00	0.41	0.48	643000.00	7.85	42.57	42.13	39.29	0.02	0.74	1.76	0.00	0.06	0.00	0.17	0.95	1301.00
N38A	255300.00	193.20	2237.00	0.11	0.05	4.56	4.64	651000.00	9.51	35.82	35.41	33.14	0.02	0.69	2.46	0.00	0.10	0.00	0.18	1.13	1832.00
N38A	237900.00	129.50	1593.00	0.09	0.07	2.90	3.00	608000.00	8.60	42.40	42.42	39.69	0.02	0.64	1.99	0.00	0.08	0.00	0.19	0.67	948.00
Total																					
Average	264810.00	89.48	2460.19	1.07	0.66	-0.01	1.41	638183.33	2.86	46.87	46.48	43.71	0.02	0.71	1.66	0.00	0.10	0.00	0.30	0.35	287.04
MAX	369500.00	1065.00	34800.00	45.10	53.10	148.00	75.30	674000.00	88.70	669.00	657.00	614.00	0.40	1.83	20.00	0.00	5.00	0.06	14.60	3.76	5360.00
MIN	190900.00	0.31	38.10	0.01	-0.19	-465.00	0.01	538000.00	0.00	0.52	0.00	0.07	0.00	0.38	-61.70	0.00	0.00	0.00	-4.94	0.00	0.16
STDEV	28070.97	124.49	3161.23	4.27	4.44	28.49	5.76	26468.95	6.98	71.23	70.64	66.33	0.03	0.22	4.17	0.00	0.34	0.00	1.14	0.49	496.51